Replacing council tax with a land value tax: a household-level microsimulation for the UK*

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Abstract

We develop a household-level microsimulation framework for land value tax reform in the United Kingdom, using PolicyEngine UK on the Enhanced Family Resources Survey, and apply it to replacing council tax — the UK's main recurrent property tax, still assessed on 1991 valuations and sharply regressive in property value. Land values are imputed to every household and benchmarked against ONS national balance sheet aggregates. We simulate a family of reform scenarios — a grid of rates from 0.5 to 5 per cent, recycling surplus revenue as a citizen's dividend, an exempt band of land value, and packages that also replace stamp duty — anchored by a budget-neutral reference scenario in which replacing £58.5 billion of net council tax revenue requires a rate of 0.79 per cent. In that scenario two thirds of households gain, absolute poverty after housing costs falls by 0.7 percentage points, and measured income inequality is essentially unchanged. The reform's incidence is organised by wealth rather than income: against the wealth ranking, council tax's Kakwani index of −0.523 becomes −0.032 — near proportionality — and the top wealth decile loses £5,871 a year, while income-ranked statistics register little. Evaluating property tax reform on income-based statistics alone misses most of what it does.

Keywords: land value tax, council tax, microsimulation, tax incidence, poverty, inequality, United Kingdom

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1 Introduction

Council tax is the United Kingdom's principal recurrent tax on residential property, with receipts forecast at approximately £54 billion for 2026–27 (OBR, 2025) and modelled gross liabilities of £62 billion (Section 3 reconciles the two). Its design has attracted sustained criticism in the public-finance literature (Mirrlees et al., 2011; Adam et al., 2020). Dwellings in England and Scotland remain assessed on their estimated value as at 1 April 1991; the banding structure is capped, with the top band in England (Band H) covering all properties worth £320,000 or more *in 1991 prices*, so that properties of widely differing current values share the top band and face nearly identical bills; and effective tax rates decline as a share of property value as that value rises. Council tax is therefore regressive with respect to the value of the asset it nominally taxes, bears a weak relationship to households' current ability to pay, and discourages investment in property improvements.

This paper develops a household-level framework for simulating land value tax reform in the UK and uses it to examine a family of scenarios in which council tax is abolished and a flat annual tax is levied on the unimproved value of land. The scenarios span a grid of tax rates from 0.5 to 5 per cent — from a net tax cut through revenue neutrality to a substantial net revenue raiser — together with packages that recycle surplus revenue as a citizen's dividend, exempt a band of land value per household, or broaden the set of taxes replaced. Throughout, we anchor the presentation on a budget-neutral reference scenario that replaces council tax's net revenue pound for pound, but this is one point in the scenario space rather than the object of the paper. The land value tax (LVT) has a long intellectual pedigree, from George (1879) through the optimal-taxation literature (Ramsey, 1927) to the review by Mirrlees et al. (2011), resting on the observation that the supply of land is fixed: a tax on pure land rents raises revenue without discouraging productive activity. The UK policy debate, however, has to date lacked a household-level quantification of the gains and losses such a reform would generate.

We provide such a quantification. Using PolicyEngine UK, an open-source static microsimulation model of the UK tax and benefit system built on an enhanced version of the Family Resources Survey (PolicyEngine, 2026), we proceed in three steps. First, we impute a land value to every household by combining survey property wealth with region-specific land shares and an allocation of corporate-sector land, benchmarking the household component against, and pinning the corporate component to, the ONS National Balance Sheet (ONS, 2025). Second, we reconstruct council tax liabilities bottom-up by region, band and discount status, calibrated to Valuation Office Agency dwelling counts. Third, we simulate the scenario family for fiscal year 2026–27; in the budget-neutral reference scenario, replacing the model's net council tax revenue of £58.5 billion (gross liabilities of £61.9 billion less £3.4 billion of council tax reduction) against an estimated UK land base of £7.44 trillion requires a rate of 0.79 per cent of land value per year; Section 3 discusses how the model-based revenue figure compares with the OBR receipts forecast.

In the budget-neutral reference scenario we report three principal findings. First, most households gain: roughly two thirds are better off, and the poorest income decile gains £584 per year, 2.7 per cent of its net income, because its average council tax bill of £1,467 substantially exceeds the £883 land tax it would owe. Second, the effect across the income distribution is otherwise close to flat and imprecisely estimated: average changes in deciles 2–8 range from −£87 to +£640 with no clear gradient, and only deciles 9 and 10 record substantial losses (£668 and £1,429). Absolute poverty after housing costs falls by 0.69 percentage points (bootstrap 95 per cent interval −1.22 to −0.20), while the changes in before-housing-cost poverty (−0.23 points) and in the Gini coefficient of equivalised net income (−0.003) are not statistically distinguishable from zero. Third, and in sharp contrast, the effect across the *wealth* distribution is monotonic and large: wealth deciles 1–7 gain between £540 and £1,453, decile 9 loses £1,111, and the top wealth decile — which holds 47 per cent of all land — loses £5,871 per year, 8.9 per cent of its net income, with 98.4 per cent of its households worse off.

The three findings are reconciled by the joint distribution of land and income. Land is far more unequally held than income (we estimate a wealth Gini of 0.70 against 0.30 for equivalised income) but is only weakly correlated with current income, so a tax on land redistributes sharply across the wealth distribution while barely registering in income-based inequality statistics. This is a substantive point about measurement as much as about land: a reform of this kind can be evaluated as approximately distribution-neutral on the statistics that conventionally govern UK tax debate, and as strongly redistributive on the dimension it actually operates on.

Across the scenario space, the rate grid from 0.5 to 5 per cent documents how revenue (from £37.2 billion to £371.8 billion), poverty and inequality vary with the rate; we decompose the tax base into household and corporate land; we verify that the swap generates no interactions with means-tested benefits, so that the household-level effect is exactly the abolished council tax bill less the land tax bill; and we test the results against alternative replacement targets, bases and geographic scope, against a proportional property tax comparator, against rent pass-through shares from zero to one, and against full capitalisation of the levy into land prices. A cross-tabulation of mean effects by income decile against wealth decile displays the paper's central dissociation directly: within every income decile, gains flip to losses only as land wealth rises.

The remainder of the paper proceeds as follows. Section 2 situates the analysis in the literature on land taxation and UK property-tax reform. Section 3 describes the data, the land-value imputation, the council tax baseline and the reform implementation. Section 4 presents the distributional results, including effective tax rates on property value, incidence by tenure, and progressivity indices computed against both income and wealth. Section 5 simulates four departures from the bare swap: recycling the proceeds of a higher rate as a citizen's dividend, an exempt band of land value, broadening the set of taxes replaced to include stamp duty and Northern Irish domestic rates, and the government's High Value Council Tax Surcharge as a comparator. Section 6 summarises the findings, discusses limitations and policy implications, and concludes. An appendix reports the full rate grid, a summary of landless households, a worked single-household example and a data and parameter summary.

2 Related literature

2.1 The economics of land value taxation

The case for taxing land rests on the classical observation that land is supplied inelastically. George (1879), in *Progress and Poverty*, argued that the rental value of unimproved land is a socially created surplus — driven by population, public infrastructure and agglomeration rather than by any effort of the owner — and proposed a "single tax" on land values to replace taxes on labour and capital. The modern optimal-taxation literature formalises the intuition: because the base cannot shrink in response to the tax, a levy on pure land rents generates no substitution effect and hence no deadweight loss, making it the canonical example of efficient taxation in the Ramsey (1927) inverse-elasticity framework. Feldstein (1977) qualified the classical view by noting general-equilibrium portfolio effects, but the presumption that land taxes are unusually efficient survives; Schwerhoff et al. (2020) survey the modern literature on rent taxation and reach the same conclusion. In subsequent work with heterogeneous households, Schwerhoff et al. (2022) show that distributional considerations can argue for taxing land value at less than 100 per cent, but that with plausible revenue recycling a land value tax reduces the net burden on low- and middle-income households — the margin our microsimulation measures directly.

Two further strands reinforce the theoretical case. Arnott and Stiglitz (1979) establish the Henry George Theorem: under conditions of optimal city size, aggregate land rents exactly finance the efficient level of local public expenditure, giving land taxation a normative role beyond efficiency — it is the natural fiscal counterpart of the public goods that generate land value. In cross-country empirical work, Johansson et al. (2008) rank recurrent taxes on immovable property as the least harmful to long-run growth among the major tax instruments, ahead of consumption, personal income and corporate income taxes. More recently, Bonnet et al. (2021) document that the post-1980 rise in wealth-to-income ratios is to a large extent a rise in the price of residential land, and argue that land is accordingly both an increasingly important and an administratively feasible tax base for contemporary economies. The measurement side of that claim rests on the residual method — valuing structures at replacement cost and attributing the remainder of property value to land — developed for the United States by Davis and Heathcote (2007), whose long-run application by Knoll et al. (2017) shows that most of the rise in house prices across advanced economies since the Second World War is a rise in land prices; the same method underlies the regional land shares we impute in Section 3.

The empirical evidence, while limited by the scarcity of implemented land taxes, is broadly consistent with the theory on the efficiency margin. Oates and Schwab (1997) examine Pittsburgh's two-rate property tax — which taxed land at more than five times the rate applied to structures — and find that the city sustained substantially higher building activity than comparable cities in the same period; they attribute this partly to the land tax's neutrality with respect to development, while noting the difficulty of isolating the tax's contribution from concurrent local conditions. Lyytikäinen (2009) finds, from Finland's three-rate property tax, that raising the tax rate on undeveloped land relative to structures increases housing construction, consistent with the development-neutrality mechanism. Tideman (1995) argues on theoretical grounds that land taxation is better than neutral, as it discourages speculative withholding of developable land. The contributions collected in Dye and England (2009) temper this optimism by documenting the practical assessment challenges — above all, separating land from improvement value at scale — that have historically limited adoption, though working national precedents exist: Denmark's *grundskyld* has taxed assessed land values separately from buildings for a century, and the Netherlands revalues every property annually under the WOZ system (OECD, 2022), which weakens the claim that separate land valuation is infeasible at scale.

The evidence on *incidence* is more contested, and it bears directly on the assumptions we maintain in Section 3. The textbook prediction is that a permanent land tax capitalises fully into land prices, so that the burden falls on owners at announcement. Høj et al. (2018) find exactly this in Danish municipal data, exploiting the 2007 local government reform, which raised land tax rates in some municipalities and lowered them in others: their estimates imply full capitalisation of the present value of future taxes at plausible discount rates. Using a later Danish quasi-experiment, however, Nielsson et al. (2024) estimate a precise *zero* effect of land taxes on residential prices, tight enough to rule out full capitalisation, and argue that the explanation lies in rent regulation: where rents are capped but tax increases may be passed through to tenants, land prices become insensitive to the tax rate and the burden is shared with tenants and future purchasers rather than borne by current owners. Beyond the Danish evidence, Löffler and Siegloch (2024) exploit more than 5,000 German municipal property tax reforms and find that around 83 per cent of a recurrent property tax — close to full pass-through — is shifted to rents within roughly three years, implying that renters may bear much of a recurrent tax on immovable property in the long run. The Danish findings are not easily reconciled, and the institutional detail that drives the second — pass-through provisions inside a rent-control regime — has no close UK counterpart in the owner-occupied sector that dominates our tax base. We nonetheless treat the incidence assumption as the least secure part of the exercise, and return to it in Sections 4.12 and 6.

2.2 UK property taxation and council tax

The deficiencies of council tax are well documented. Mirrlees et al. (2011) concluded that UK housing taxation lacks a coherent rationale: council tax is regressive in property value and based on obsolete 1991 valuations, while stamp duty land tax taxes transactions and thereby impedes mobility. The second half of that verdict has since been quantified: Best and Kleven (2018) show, using the universe of UK property transactions and quasi-experimental variation from the pre-2014 slab notches, that transaction taxes distort the price, volume and timing of sales substantially, and Hilber and Lyytikäinen (2017) find that a 2 percentage point increase in stamp duty reduces annual mobility by roughly 30 per cent, with the distortion concentrated in housing-related rather than job-related moves. A recurrent tax on an immobile base avoids this margin entirely, which is a large part of the case for the swap we simulate. The review proposed replacing both with a Housing Services Tax proportional to the consumption value of housing, complemented by a land value tax on business land — an explicit endorsement of land taxation where valuation is feasible. Subsequent work has quantified the regressivity that motivates reform: Adam et al. (2020) show that effective council tax rates as a share of property value fall sharply across the price distribution, and that the absence of revaluation in England and Scotland leaves relative price changes since 1991 — most notably London's appreciation — untaxed. Muellbauer (2005) argues that reforming council tax into a proportional property tax with regular automatic revaluation would also stabilise the housing market, and the OECD's most recent survey of the UK likewise recommends updating property valuations and reforming council tax (OECD, 2024).

A body of applied UK work has modelled alternatives. Corlett and Gardiner (2018) examine options including a proportional property tax and a partial LVT at the aggregate and regional level; Fairer Share (2021) propose a proportional property tax of 0.48 per cent of current property value and estimate that roughly three-quarters of households would pay less — a useful comparator for our finding that 67 per cent of households gain at a 0.79 per cent tax on the narrower land base. Murphy and Snelling (2019) model council tax reform for London. In Scotland, the cross-party Commission on Local Tax Reform (Scottish Government, 2015) concluded that the present system must end and assessed property, land and income bases, and Hughes et al. (2018) examine LVT options for the Scottish Land Commission, highlighting valuation and transition as the binding constraints. Leunig (2024) proposes replacing both council tax and stamp duty with proportional levies on current values. Most recently, IFS (2025) conduct for the Scottish Government a household-level analysis of five revaluation and reform options, finding that a pure revenue-neutral revaluation would raise average net bills in the top income quintile by £267 a year and reduce them across the rest of the distribution — a useful benchmark for the magnitude of what revaluation alone achieves, against which our land-based swap can be read.

Closest to our exercise, ap Gwilym et al. (2020) assess the feasibility of a local land value tax for Wales for the Welsh Government, simulating revenue-neutral rates at local-authority level using small-area land value estimates. In work concurrent with ours, Neidle (2026) builds a property-level model of the same replacement from Land Registry transaction data, estimating a revenue-neutral rate of 1.28 per cent for replacing council tax and stamp duty together, with roughly 69 per cent of households paying less, and explores progressive rates, regional variants and transition design; and the parcel-level Land Value Tax Calculator (Land Value Tax Calculator, 2026) maps estimated land values for individual properties across England and Wales. Both are complementary to the household-level approach taken here: transaction- and parcel-based models resolve spatial detail our survey base cannot, while microsimulation inside the full tax-benefit system resolves household incidence, benefit interactions and poverty effects; our stamp duty package in Section 5 provides the directly comparable scenario.

Policy has meanwhile moved. At the Autumn Budget 2025 the UK government announced a High Value Council Tax Surcharge (HM Treasury, 2025): from April 2028, English residential properties worth £2 million or more, valued by the Valuation Office at 2026 prices and revalued every five years, will pay a flat annual surcharge of £2,500 (£2–2.5m), £3,500 (£2.5–3.5m), £5,000 (£3.5–5m) or £7,500 (above £5m), raising approximately £0.4 billion a year from around 165,000 properties. This is instructive in two ways. Substantively, it concedes the central criticism of council tax — that the banding structure stops discriminating at the top — while addressing it through a bolt-on surcharge on a fraction of one per cent of the stock rather than through revaluation, and it raises around 0.7 per cent of the revenue we consider reallocating. Administratively, it establishes that a targeted revaluation of high-value English homes at current prices is regarded as feasible by the tax authority, which weakens the standard objection that up-to-date valuation is impracticable. It takes effect in 2028 and so does not enter our 2026–27 baseline.

2.3 Contribution

Relative to this literature, our contribution is a household-level microsimulation of a full council-tax-to-LVT swap for the UK, with land values imputed to every household in a nationally representative micro-dataset, benchmarked against national balance-sheet aggregates, and evaluated within a complete model of the tax and benefit system. The exercise sits squarely in the tax-benefit microsimulation tradition exemplified by EUROMOD (Sutherland and Figari, 2013), extended from income flows to a wealth-stock base; the closest UK methodological precedent is Advani et al. (2021), who use the Wealth and Assets Survey to model the revenue and distributional effects of an annual wealth tax, and we follow Goedemé et al. (2013) in treating survey-based point estimates as subject to sampling error rather than exact. Prior UK LVT analyses have generally worked with regional, band-level or local-authority aggregates — the Welsh assessment of ap Gwilym et al. (2020) being the most detailed to date — rather than with household microdata. The recent household-level work that does exist takes the property base as given and asks how to band or revalue it (IFS, 2025; Adam et al., 2020), or targets the top of the distribution with a surcharge (HM Treasury, 2025); we instead change the base itself, from dwellings to land, and hold revenue fixed. The microdata buy us three things that aggregate analyses cannot deliver: the joint distribution of land and income, which drives the winners-and-losers results and the divergence between income-ranked and wealth-ranked incidence; poverty rates measured against HBAI-style equivalised, housing-cost-adjusted thresholds; and a decomposition of who bears the corporate component of the base. We note in Section 4 that the swap turns out to generate no interactions with means-tested benefit entitlements, so the household-level net income change is exactly the difference between the abolished council tax bill and the new land tax bill — a verified property of our modelled implementation, which does not enter the levy into means-tested income or capital tests, and one that materially simplifies interpretation. To our knowledge this is the first open-source, household-level distributional analysis of a revenue-neutral LVT for the whole of the UK, and the code is open source and the analysis reproducible conditional on the released model and licensed microdata (PolicyEngine, 2026).

3 Data and methodology

3.1 Microdata

Our simulations use PolicyEngine UK (PolicyEngine, 2026), an open-source static microsimulation model of the UK tax and benefit system, run on the Enhanced Family Resources Survey (EFRS) for 2023–24, uprated to fiscal year 2026–27. The EFRS augments the standard Family Resources Survey in two ways. First, because household surveys under-represent the top of the income distribution, synthetic high-income records are generated by training a quantile random forest on HMRC's Survey of Personal Incomes (SPI) and imputing the upper tail of incomes. Second, household weights are calibrated by gradient descent so that the weighted dataset matches a large set of national and regional administrative totals from the ONS, HMRC and DWP — including population by age and region, income components, and benefit caseloads. Wealth variables, including property wealth and corporate (financial) wealth, are imputed from the ONS Wealth and Assets Survey (WAS). These enhancement steps are implemented in the upstream PolicyEngine UK data pipeline, which we take as given; the code released with this paper consumes the published EFRS and implements the land imputation, reform scenarios and analysis.

3.2 Council tax baseline

The FRS records council tax imperfectly, so we reconstruct liabilities. Each household is assigned a council tax band (A–H in England and Scotland, A–I in Wales) and a bill equal to the average band rate in its region, with the 25 per cent single-person discount imputed to single-adult households. Band assignment probabilities are calibrated so that the distribution of dwellings by region and band matches Valuation Office Agency (and Scottish Assessors) counts. This bottom-up reconstruction yields, for 2026–27, gross council tax of £61.9 billion; netting off £3.4 billion of means-tested council tax reduction gives net revenue of

$$R_{\text{CT}} \;=\; 61.9 - 3.4 \;=\; \pounds 58.5 \text{ billion}, \tag{1}$$

which is the amount the land value tax must replace.

We emphasise that this is a modelled liability total, calibrated to dwelling counts by region and band, and not a receipts figure. It exceeds the Office for Budget Responsibility's accruals-based council tax receipts forecast of approximately £53.7 billion (OBR, 2025) by some 9 per cent of net revenue. Part of the gap is conceptual — receipts are net of collection losses, discounts and exemptions that a liability model applied to average band rates does not fully reproduce — and part reflects genuine calibration error in the band-rate averages. The overshoot is not evenly spread across nations: modelled English liabilities are within a few per cent of the English council tax requirement, whereas modelled Scottish (£4.0 billion net) and Welsh (£3.1 billion) liabilities exceed the corresponding devolved budget figures by roughly a fifth. Because the swap credits each household with its abolished bill, any overstatement of Scottish and Welsh council tax translates one-for-one into overstated gains for Scottish and Welsh households, a point that matters for the geographic discussion in Section 4.9. Because the headline rate is the ratio of this figure to the land base, we report throughout a second replacement target calibrated to OBR receipts, which implies a revenue-neutral rate of 0.72 rather than 0.79 per cent; Section 4.12 reports the distributional results under both. The 0.79 per cent central case has the advantage of internal consistency: the simulated LVT replaces exactly the council tax the model abolishes, household by household, so that aggregate net income is unchanged by construction. Northern Ireland operates a separate domestic rates system rather than council tax; our central specification leaves it unchanged, and Section 4.12 reports results excluding Northern Ireland.

3.3 Imputing land values

The tax base is the unimproved value of land held directly (residential land under owner-occupied and other household-owned property, plus any land held as an asset in its own right) and indirectly (land on corporate balance sheets, attributed to households as ultimate owners). For household i in region r, land value is

$$L_i = \underbrace{\lambda_r P_i + A_i}_{\text{household land } H_i} + \underbrace{\frac{K_i}{\sum_j K_j w_j} L^{\text{corp}}}_{\text{allocated corporate land } C_i}, \tag{2}$$

where P_i is WAS-imputed property wealth, λ_r is the region-specific land share of property value, A_i is directly owned land reported separately in the WAS asset schedule, K_i is the household's corporate wealth, w_j are survey weights, and L^{corp} is aggregate corporate-sector land. Corporate land is allocated proportionally to each household's corporate wealth, treating shareholdings (direct and via pensions) as the ultimate claim on corporate assets. Note that the allocation is normalised to the weighted population total, so $\sum_i C_i w_i = L^{\text{corp}}$ holds exactly by construction; the corporate component of the base is therefore pinned to its aggregate, whereas the household component is not.

Regional land shares. The shares λ_r are derived by the residual method from the MHCLG *Land Value Estimates for Policy Appraisal* (MHCLG, 2026) and the UK House Price Index; the residual approach — valuing structures at replacement cost and attributing the remainder of property value to land — originates with Davis and Heathcote (2007) and underlies the long-run house price decompositions of Knoll et al. (2017). MHCLG publishes these figures as generic estimates for policy appraisal — typical serviced sites under standardised assumptions — and cautions that they do not represent market values; we therefore treat the resulting λ_r as appraisal-based proxies for market land shares, a caveat that bears on the level of the base more than on its distribution, and one the share perturbations in Section 4.12 speak to. For each English local authority we take the MHCLG central-scenario residential land value per hectare, divide by dwellings per hectare to obtain land value per dwelling, and divide that by the average house price, giving

$$\lambda_r = \frac{\text{MHCLG land value per hectare}_r \,/\, \text{dwellings per hectare}_r}{\text{average house price}_r}, \tag{3}$$

aggregated from local authority to region. Because MHCLG covers England only, Scotland, Wales and Northern Ireland are interpolated from the English regions closest in house price (Scotland and Northern Ireland from the North East and North West, Wales from the North West). Table 1 reports the resulting values. They range from 85 per cent in London to 42 per cent in the North East, reflecting the fact that in high-price regions almost all of the price differential is location value rather than structure cost, while in low-price regions the price approaches the cost of the structure. The interpolation for the devolved nations is a material assumption — roughly 13 per cent of GB dwellings are assigned a land share estimated from English data — and we test sensitivity to it in Section 4.12. The Northern Ireland assignment deserves a separate flag: an earlier release assigned it the population-weighted mean of English shares (0.67), which sits above every English region outside London and the South East — implausible given that Northern Ireland has roughly the lowest house prices in the UK, since by the logic of the residual method its share should be near the bottom of the range. Our central specification therefore interpolates Northern Ireland like Scotland and Wales, from the price-similar North East and North West (0.44); Section 4.12 reports a variant using the superseded 0.67 share, under which the aggregate results barely move but Northern Irish liabilities rise by roughly a half, which matters for the geographic discussion in Section 4.9.

Table 1: Regional land share of property value, λ_r

Region	λ_r	Basis
London	0.85	MHCLG, England
South East	0.65	MHCLG, England
East of England	0.60	MHCLG, England
South West	0.58	MHCLG, England
West Midlands	0.52	MHCLG, England
East Midlands	0.48	MHCLG, England
North West	0.47	MHCLG, England
Wales	0.47	Interpolated (North West)
Yorkshire and the Humber	0.46	MHCLG, England
Scotland	0.44	Interpolated (North East / North West)
North East	0.42	MHCLG, England
Northern Ireland	0.44	Interpolated (North East / North West)

Source: Equation (3), MHCLG *Land Value Estimates for Policy Appraisal 2023* central scenario and UK House Price Index; implemented in PolicyEngine UK as `household.wealth.land.intensity.property_wealth_by_region`.

Aggregates. The corporate aggregate L^{corp} is taken from the sectoral tables underlying the ONS National Balance Sheet (ONS, 2025), which give UK land of £7.10 trillion at end-2024, split £5.04 trillion household and £2.06 trillion corporate. (These derived series differ from the headline bulletin presentation, which reports household non-produced assets of approximately £4.6 trillion in 2024 on a different sectoral grouping; the household/corporate split is applied using the last directly observed sectoral shares, from 2020.) The corporate series is held at its 2024 level for 2025 and 2026, so $L^{\text{corp}} = £2.06$ trillion enters the simulation unchanged.

The directly owned component A_i deserves a flag of its own: it aggregates to £0.67 trillion, 12 per cent of household land, and it is both unevenly distributed across nations — markedly more important in Scotland and Wales than in England on this data release — and unstable across releases of the underlying WAS imputation. It is survey-reported land held as an asset in its own right, dominated by a small number of large agricultural holdings, and its imputation into the FRS is correspondingly noisy. Section 4.12 reports the results with A_i excluded from the base entirely.

The household aggregate is *not* imposed. It emerges from equation (2) applied to EFRS property wealth, which is itself uprated from the 2023–24 survey to 2026–27 by the model's house-price and financial-asset uprating indices. The weighted household total is

$$H \;=\; \sum_i H_i w_i \;=\; \pounds 5.38 \text{ trillion}, \tag{4}$$

which is 106.7 per cent of the corresponding un-uprated 2024 ONS household benchmark of £5.04 trillion. Roughly speaking, the 7 per cent excess is the two years of nominal property-price growth embodied in the uprated microdata, with the residual reflecting imputation error in WAS property wealth and the fact that the regional household-land targets are not binding constraints in the weight calibration. The total base is therefore

$$L \;=\; H + L^{\text{corp}} \;=\; 5.38 + 2.06 \;=\; \pounds 7.44 \text{ trillion}, \tag{5}$$

or 104.8 per cent of the un-uprated 2024 ONS total. We report this as a model output benchmarked against ONS rather than as a calibrated target, and Section 4.12 shows how the results move if the household base is instead scaled to the ONS 2024 level. Household land so measured is 68.0 per cent of total property wealth and total land is 27.6 per cent of total household wealth (£26.98 trillion); mean land value per household, including allocated corporate land, is £231,833 and the median is £93,516.

3.4 The reform scenarios and the reference rate

Every scenario we simulate (i) abolishes council tax in Great Britain and (ii) levies a flat annual tax at rate r on each household's land value L_i, implemented in PolicyEngine UK via the parameters `gov.contrib.ubi_center.land_value_tax.rate` and `gov.contrib.abolish_council_tax`; scenarios differ in the rate r, in what is done with any revenue beyond the council tax replacement (Section 5.1), in exemptions (Section 5.2), and in the set of taxes replaced (Section 5.3). The budget-neutral reference rate equates LVT revenue with net council tax revenue:

$$r^* \;=\; \frac{R_{\text{CT}}}{L} \;=\; \frac{58.5}{7{,}436} \;=\; 0.79\%. \tag{6}$$

The net income change for household i is $\Delta y_i = \text{CT}_i - r^* L_i$, where CT_i is the abolished council tax liability (net of council tax reduction). Three maintained assumptions govern incidence, all standard in static microsimulation. First, incidence is assigned entirely to current landowners: we assume no behavioural response and no capitalisation of the tax into land prices. Second, rents are held fixed, so renting households with no property holdings retain their abolished council tax liability in full and pay LVT only on any corporate wealth they hold (a renting household that owns let or second property is a landowner on that margin). This second assumption is the one most exposed to the empirical literature: Nielsson et al. (2024) find that Danish land taxes are passed through to tenants where rent regulation permits it, and Löffler and Siegloch

(2024) find close-to-full pass-through (around 83 per cent) of German recurrent property taxes to rents within roughly three years — the strongest evidence against the fixed-rent case — in which case renters would bear part of the levy and gain less than we report. The UK private rented sector has no equivalent statutory pass-through mechanism, and the residential component of our base is dominated by owner-occupied land, but the assumption should be read as maintained rather than established. Section 4.12 reports both a household-land-only variant, which brackets how much of the result depends on the non-owner-occupied part of the base, and an explicit pass-through sensitivity that reports incidence as a function of the share of the levy passed through to rents. Third, the land base is held fixed across the rate grid, consistent with the inelastic supply of land. All results are for simulation year 2026. Statistical conventions follow official practice: poverty is measured as absolute poverty against a baseline-fixed threshold, before housing costs (BHC) and after housing costs (AHC), and is reported as the share of *individuals* living in poor households (the HBAI convention); income inequality is the Gini coefficient of equivalised household net income, person-weighted; wealth inequality is the Gini of total household wealth, household-weighted (the ONS Wealth and Assets Survey convention). A household counts as a winner (loser) if its net income rises (falls) by more than £1 a year; changes within £1 of zero are reported as unchanged. The reference rate $r^* = 0.79\%$ is one point on a grid of rates $\{0.5, 1.0, 1.5, 2.0, 3.0, 5.0\}$ per cent that we simulate throughout to trace out revenue and distributional effects; we also decompose revenue by base scope (household-only versus corporate-only land).

4 Results

4.1 Effective tax rates on property value

Before turning to the distribution of the base, it is worth showing the defect the reform is meant to correct. Table 2 reports council tax and the budget-neutral LVT as percentages of property value, for the 61.5 per cent of households that own residential property.[1] Council tax falls from 1.78 per cent of value for properties under £150,000 to 0.07 per cent for properties above £2 million: a household in the most valuable band pays 25 times less, as a share of what it owns, than one in the least valuable band, and in cash terms it pays *less* in absolute terms (£2,554 against £2,582 in the £1–2 million band) — the mean bill is essentially flat at the top, consistent with the capped band structure. This is the regressivity documented by Adam et al. (2020), and it is the direct consequence of capping bands at 1991 values.

The land value tax is close to flat on the same axis, ranging between 0.42 and 0.67 per cent of property value with no systematic gradient. It is not exactly flat, and the reason is instructive: the ratio reflects the regional land share λ_r and the household's corporate wealth, not the value of the dwelling, so it varies with where a property is rather than with what it is worth. The reform therefore does not simply make the existing tax proportional — it replaces a charge that

[1] The value concept here is *all* residential property held by the household, including second and let homes, which is the base the household-land imputation runs on. Counts at the top of this distribution are therefore not comparable with counts of main residences: 664,000 households hold residential property worth over £2 million on this definition, but only 183,000 own a *main residence* worth that much (171,000 in England), which reconciles with the roughly 165,000 properties the OBR expects to be liable for the High Value Council Tax Surcharge (Section 5.4). The non-monotonic band shares in the upper tail (3.0 per cent at £750k–1m against 7.1 per cent at £1–2m) reflect the WAS/SPI imputation of the upper tail rather than genuine structure, and band shares may not sum exactly to the ownership rate because of rounding.

is sharply regressive in property value with one that is approximately proportional to it and exactly proportional to land.

Table 2: Council tax and LVT as a share of property value, property owners

Property value band	Share of households (%)	Avg. property value (£)	Avg. council tax (£)	As % of property value	
				Council tax	LVT
Under £150k	15.3	97,214	1,727	1.78	0.60
£150–200k	6.4	174,849	2,077	1.19	0.62
£200–300k	10.0	244,157	2,208	0.90	0.67
£300–400k	6.5	352,256	2,371	0.67	0.60
£400–500k	6.1	450,709	2,421	0.54	0.62
£500–750k	5.2	590,818	2,427	0.41	0.65
£750k–1m	3.0	873,421	2,622	0.30	0.56
£1–2m	7.1	1,414,114	2,582	0.18	0.49
Over £2m	2.1	3,589,816	2,554	0.07	0.42
Memo: no property	38.5	—	1,209	—	—

Note: Households owning no residential property are excluded from the rate columns, since council tax over a property value of zero is not an effective rate; they are shown as a memo row and pay LVT only on attributed corporate land (£138 on average). *Source:* Authors' calculations; `analysis/deep_dive.py`.

4.2 Where land value sits

The distributional consequences of an LVT are determined by the distribution of its base, which we describe first. Table 3 reports average land value, property wealth and land shares by decile of equivalised household net income. Two features stand out. First, average land value rises with income, but weakly and unevenly: it is £112,355 in the poorest decile and £497,218 in the richest, a ratio of 4.4 to 1 against an income ratio of 6.5 to 1, and the profile is far from monotonic — decile 2 (£204,538) holds about twice as much land on average as decile 3 (£100,901). Some of this reflects genuine heterogeneity, principally asset-rich retired households with low current income; some reflects imputation noise in a base constructed from survey property wealth. Either way, the income gradient of the tax base is shallow, which is what drives the results in Section 4.10. Second, the top two income deciles together hold 35.9 per cent of land, whereas, as Section 4.5 shows, the top two *wealth* deciles hold 66.7 per cent.

Table 3 separates the two components of the base, which are not comparable quantities: household land is a share of the household's own property wealth, while allocated corporate land is an attributed claim on corporate balance sheets that households holding no property may still bear. Reporting them together against property wealth — as a single "land value" column would — can make land appear to exceed the property wealth it is derived from.

Regional variation is at least as pronounced as variation by income. Average household land value is £369,802 in London and £286,311 in the South East, compared with £153,631 in the North East and £159,535 in Scotland (Figure 2) — a ratio of 2.4 to 1 that reflects both higher property prices and higher land shares of those prices in the South. Across household types, pensioner couples hold the most land on average (£431,211), followed by working-age couples without children (£302,350); lone parents, who are predominantly renters, hold very

Table 3: Land value and property wealth by decile of equivalised household net income, 2026–27

Income decile	Avg. income (£/yr)	Avg. household land (£)	Avg. corporate land (£)	Avg. total land (£)	Avg. property wealth (£)	Share of land (%)
1	21,391	86,284	26,071	112,355	123,508	5.6
2	31,942	165,548	38,990	204,538	190,189	8.1
3	36,983	77,422	23,480	100,901	124,226	4.3
4	38,540	138,699	41,941	180,640	181,346	8.4
5	43,217	163,955	62,907	226,863	245,660	10.4
6	54,707	162,225	53,141	215,366	239,736	8.5
7	55,218	143,075	47,821	190,896	224,979	8.3
8	68,606	174,345	81,906	256,250	288,363	10.2
9	78,412	247,985	120,878	368,863	404,732	15.1
10	138,195	340,804	156,414	497,218	479,920	20.8

Source: Authors' calculations using PolicyEngine UK.

little (£7,790) (Figure 3). Ranked by total wealth rather than income, concentration is far greater: the bottom three wealth deciles hold essentially no land, while the top wealth decile averages £1,085,567 — 47.5 per cent of all land (Table 7 below). Overall, 17.2 per cent of households hold no land.

4.3 Revenue and the rate schedule

Because the base is fixed in the static framework, LVT revenue is linear in the rate: each 0.1 percentage point raises approximately £7.4 billion. Table 4 reports revenue across the rate grid. A 0.5 per cent rate raises £37.2 billion — £21.3 billion less than council tax — while a 1 per cent rate raises £74.4 billion and a 5 per cent rate raises £371.8 billion. The budget-neutral rate is 0.79 per cent (equation 6). Decomposing the base at a 1 per cent rate, household land contributes £53.8 billion and corporate land £20.6 billion of the £74.4 billion total. The corporate component — 28 per cent of revenue — falls on households in proportion to their shareholdings, and its inclusion permits a lower rate for any given revenue target than a purely residential base would require.

Table 4: LVT revenue by rate, 2026–27

Rate	LVT revenue (£bn)	Council tax (£bn)	Net revenue (£bn)	Avg. LVT per household (£)
0.5%	37.2	58.5	–21.3	1,159
1.0%	74.4	58.5	15.9	2,318
1.5%	111.5	58.5	53.1	3,478
2.0%	148.7	58.5	90.3	4,637
3.0%	223.1	58.5	164.6	6,955
5.0%	371.8	58.5	313.4	11,592

Note: Net revenue is LVT revenue less replaced council tax revenue. *Source:* Authors' calculations using PolicyEngine UK.

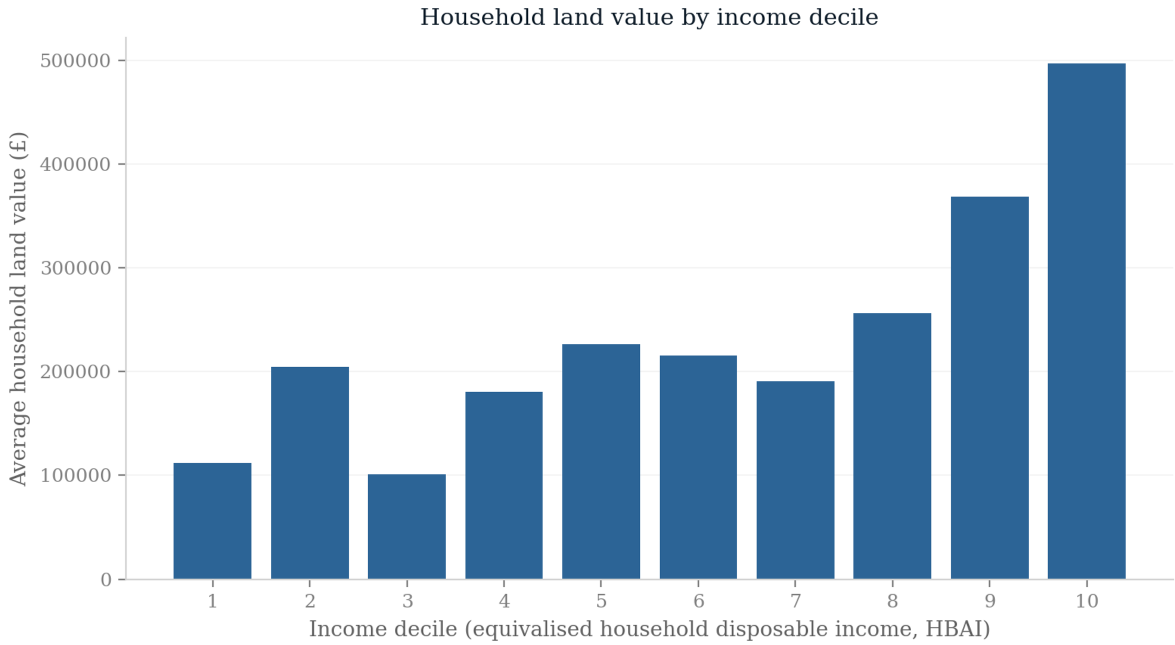

Figure 1: Average land value by income decile.

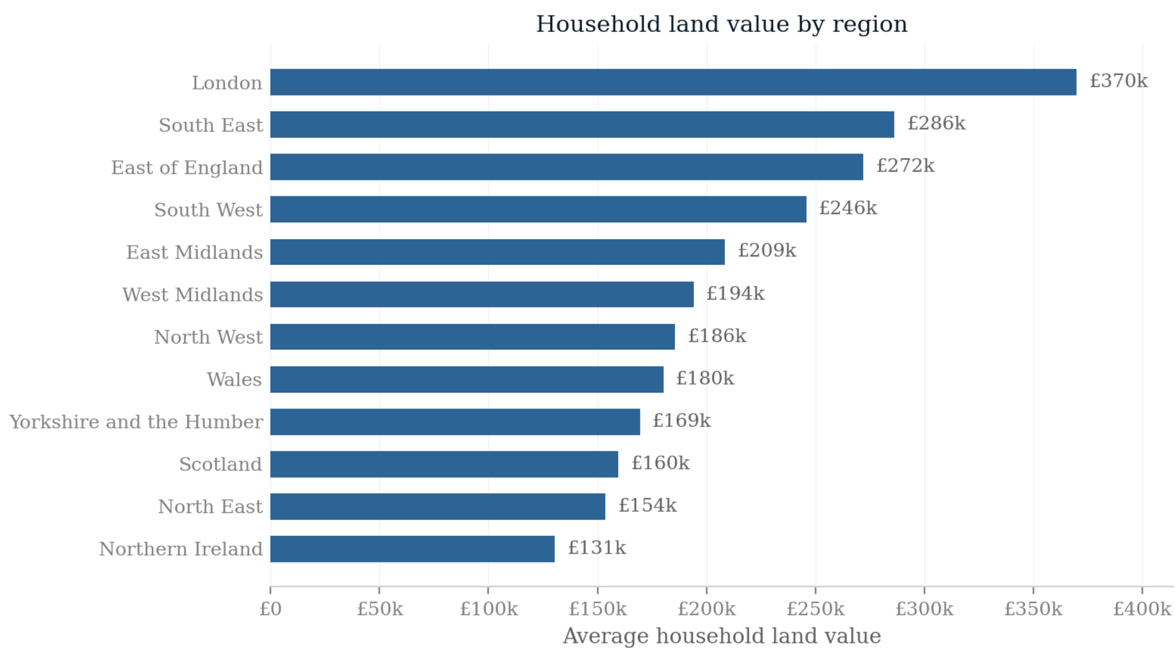

Figure 2: Average total land attributed per household, including corporate land, by region.

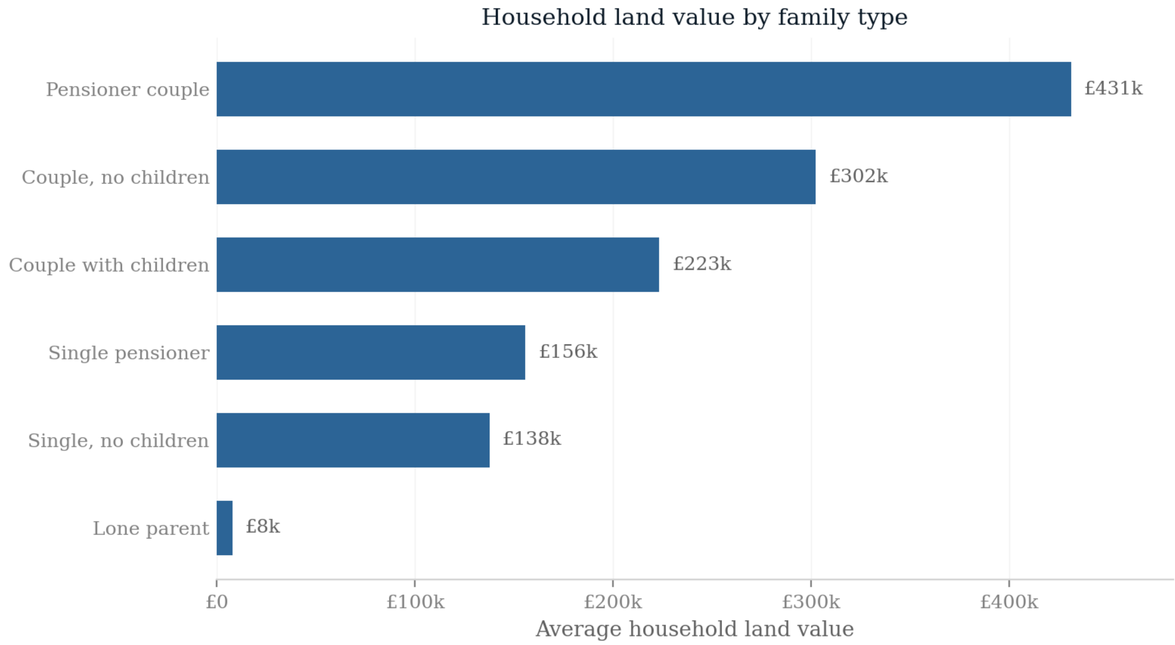

Figure 3: Average total land attributed per household, including corporate land, by family type.

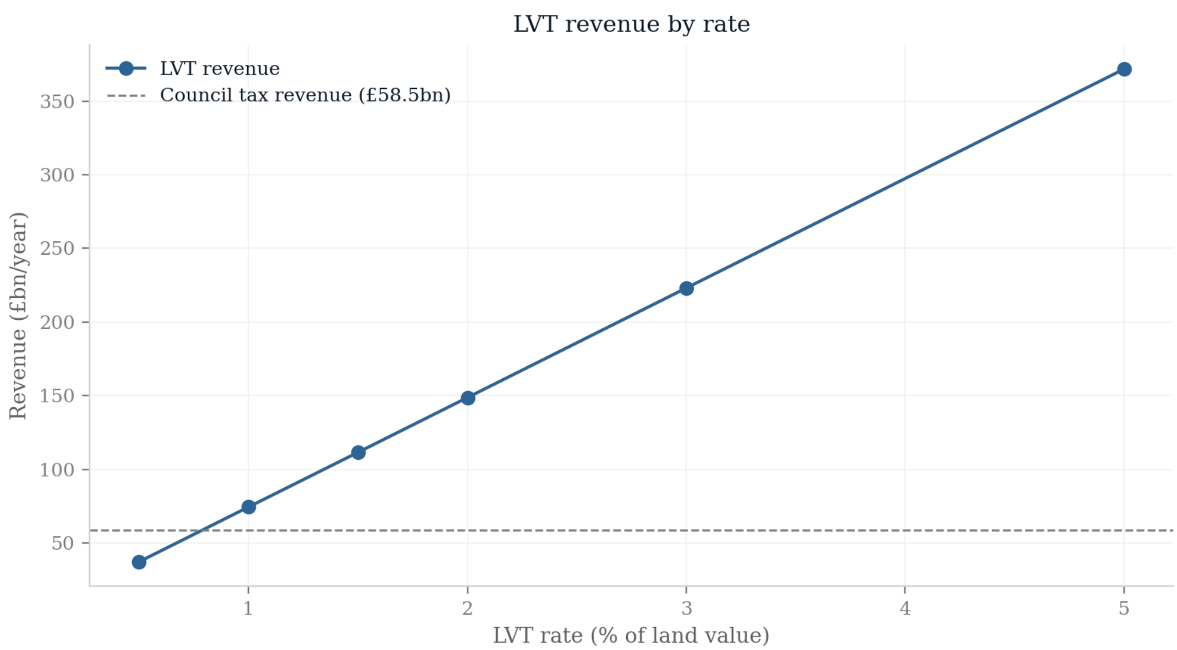

Figure 4: LVT revenue by rate against the council tax replacement target.

4.4 Distributional impact by income decile

Table 6 and Figure 5 report the average change in household net income at the budget-neutral 0.79 per cent rate, by income decile. Because the swap generates no benefit interactions — a result we verify in Section 4.12 — the change is exactly the abolished council tax bill less the land tax bill, $\Delta y_i = \mathrm{CT}_i - r^* L_i$.

The pattern across income deciles is best described as flat with a step down at the top. The poorest decile gains most in both absolute and relative terms, £584 per year or 2.7 per cent of net income, because its average council tax bill of £1,467 — inflated relative to its means by the compressed banding structure — is well above the £883 land tax charged on its modest holdings. Deciles 3 and 7 gain £640 and £385; deciles 4, 6 and 8 record small average gains of £122–£173; and deciles 2 and 5 record small average *losses* of £17 and £87, reflecting the uneven land profile documented in Table 3 rather than anything systematic about those income levels. Only in deciles 9 and 10 is the loss substantial and clearly attributable to income: £668 and £1,429 per year, as land holdings of £369,000–£497,000 attract liabilities of £2,900–£3,910 against council tax savings of £2,230–£2,480. In percentage terms the losses are similar (−0.9 per cent for decile 9 and −1.0 per cent for decile 10), because top-decile incomes are considerably higher while average land holdings are only moderately larger.

Table 5 attaches bootstrap confidence intervals to these averages, resampling households with replacement 500 times. They confirm what the point estimates suggest: only five of the ten decile means are distinguishable from zero at the 5 per cent level. Deciles 1, 3 and 7 are significantly positive, deciles 9 and 10 significantly negative, and deciles 2, 4, 5, 6 and 8 — the entire middle of the distribution — are not significantly different from zero. Decile 4's interval runs from −£954 to +£775. The apparent zig-zag across the middle deciles is therefore consistent with sampling and imputation noise rather than structure, on intervals that reflect weight resampling only.

Table 5: Average net income change by income decile with bootstrap confidence intervals

Income decile	Avg. net change (£)	95% interval Low	95% interval High	Distinguishable from zero
1	+584	+277	+814	yes
2	−17	−742	+587	no
3	+640	+335	+873	yes
4	+122	−954	+775	no
5	−87	−642	+382	no
6	+173	−720	+651	no
7	+385	+7	+734	yes
8	+161	−145	+414	no
9	−668	−1,288	−171	yes
10	−1,429	−2,220	−717	yes

Note: 500 bootstrap draws, households resampled with replacement, deciles held at their full-sample definition. The intervals capture sampling variability only: the WAS-to-FRS wealth imputation, the regional land shares and the survey's clustered design are held fixed across draws, so total uncertainty is understated. *Source:* Authors' calculations; `analysis/deep_dive.py`.

We stress that the middle of this profile is not precisely estimated. The differences between adjacent deciles in the range 2–8 are of the order of a few hundred pounds a year on average net incomes of £32,000–£69,000, and they rest on imputed land values whose decile means are visibly noisy. The result that should be read from Table 6 is that the reform is roughly distributionally neutral across the bottom eight deciles taken together and clearly costly to the top two — not that any particular middle decile gains or loses.

Table 6: Impact of the 0.79% budget-neutral LVT by income decile

Income decile	Avg. LVT (£)	Avg. CT saved (£)	Avg. net change (£)	% of net income	Winners (%)	Losers (%)
1	883	1,467	+584	+2.7	77.9	18.2
2	1,608	1,591	−17	−0.1	76.9	21.4
3	793	1,434	+640	+1.7	71.6	18.5
4	1,420	1,542	+122	+0.3	72.1	25.8
5	1,783	1,696	−87	−0.2	63.8	33.3
6	1,693	1,866	+173	+0.3	69.7	27.4
7	1,501	1,885	+385	+0.7	75.1	24.1
8	2,014	2,175	+161	+0.2	70.4	27.9
9	2,899	2,231	−668	−0.9	48.4	50.9
10	3,908	2,479	−1,429	−1.0	45.4	53.9

Note: "CT saved" denotes the abolished council tax liability net of council tax reduction. A winner (loser) is a household whose net income changes by more than ±£1 a year; winners and losers do not sum to 100 per cent because the remainder are unaffected. *Source:* Authors' calculations using PolicyEngine UK.

4.5 Distributional impact by wealth decile

Table 7 and Figure 6 report the corresponding results when households are ranked by total wealth rather than income.[2] The gradient here is monotonic and steep, in sharp contrast to the income ranking. The bottom two wealth deciles own essentially no land and retain their council tax savings in full, gaining £952 and £959 per year; deciles 3–7 gain between £540 and £1,453; decile 8 loses £261; decile 9 loses £1,111; and the top wealth decile — averaging £1.09 million of land — loses £5,871 per year, or 8.9 per cent of its net income, with an average LVT liability of £8,533 against council tax savings of £2,662. In the top wealth decile 98.4 per cent of households lose; in each of the bottom seven, roughly 80 per cent or more gain. The incidence of the reform tracks land wealth almost mechanically, which is exactly what the instrument is designed to do.[3]

[2] We construct equal-weight wealth deciles directly. PolicyEngine UK's own `household_wealth_decile` variable is degenerate at the bottom of the distribution — the mass of households with zero or negative net wealth is tied, leaving decile one empty and placing 22.6 per cent of households in decile two — which would make the decile averages in this table incomparable.

[3] Revenue neutrality holds in aggregate — LVT revenue equals abolished council tax revenue at £58.5 billion — so the population-weighted average net-income change is zero. It is identical whether households are grouped by income or by wealth decile, since the two tables show the same households re-sorted; the per-decile averages need not sum to zero because decile group sizes are equal in population but not in the underlying quantities.

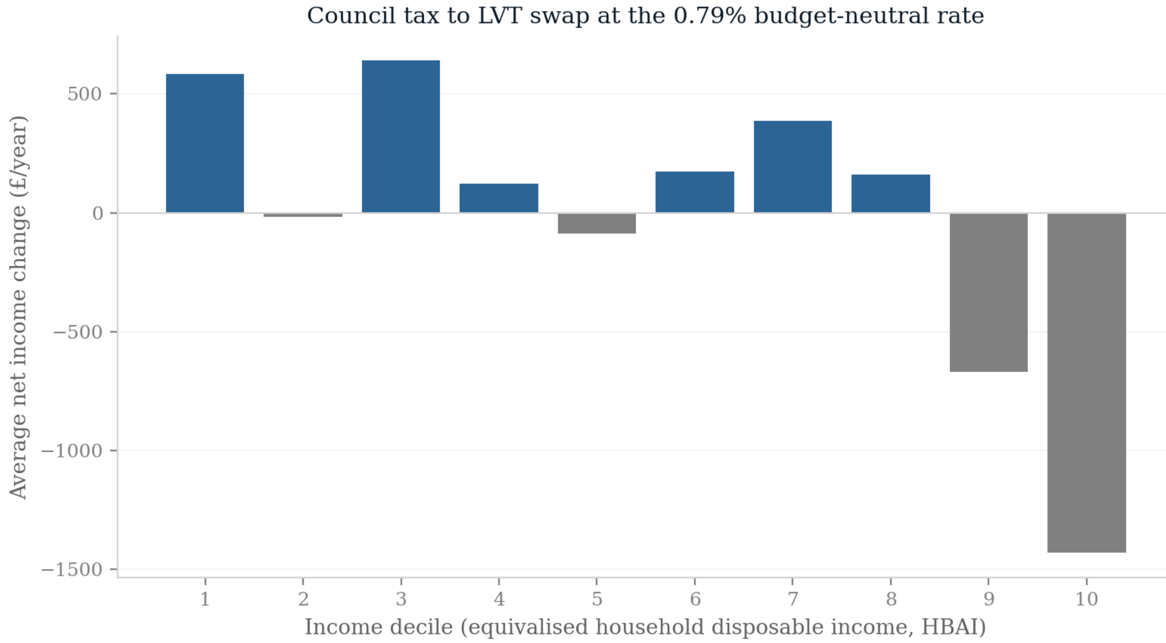

Figure 5: Average net income change by income decile at the 0.79% budget-neutral rate.

Table 7: Impact of the 0.79% budget-neutral LVT by wealth decile

Wealth decile	Avg. land value (£)	Avg. LVT (£)	Avg. CT saved (£)	Avg. net change (£)	Winners (%)	Losers (%)
1	1	0	952	+952	79.4	9.9
2	29	0	959	+959	83.2	6.9
3	1,265	10	1,215	+1,205	83.8	9.4
4	25,099	197	1,650	+1,453	92.1	7.9
5	76,396	600	1,858	+1,258	94.1	5.9
6	138,545	1,089	2,041	+952	89.4	10.6
7	211,325	1,661	2,201	+540	82.9	17.0
8	320,260	2,517	2,256	−261	48.3	51.6
9	449,504	3,533	2,422	−1,111	19.5	80.5
10	1,085,567	8,533	2,662	−5,871	1.6	98.4

Note: Deciles of total household wealth. *Source:* Authors' calculations using PolicyEngine UK.

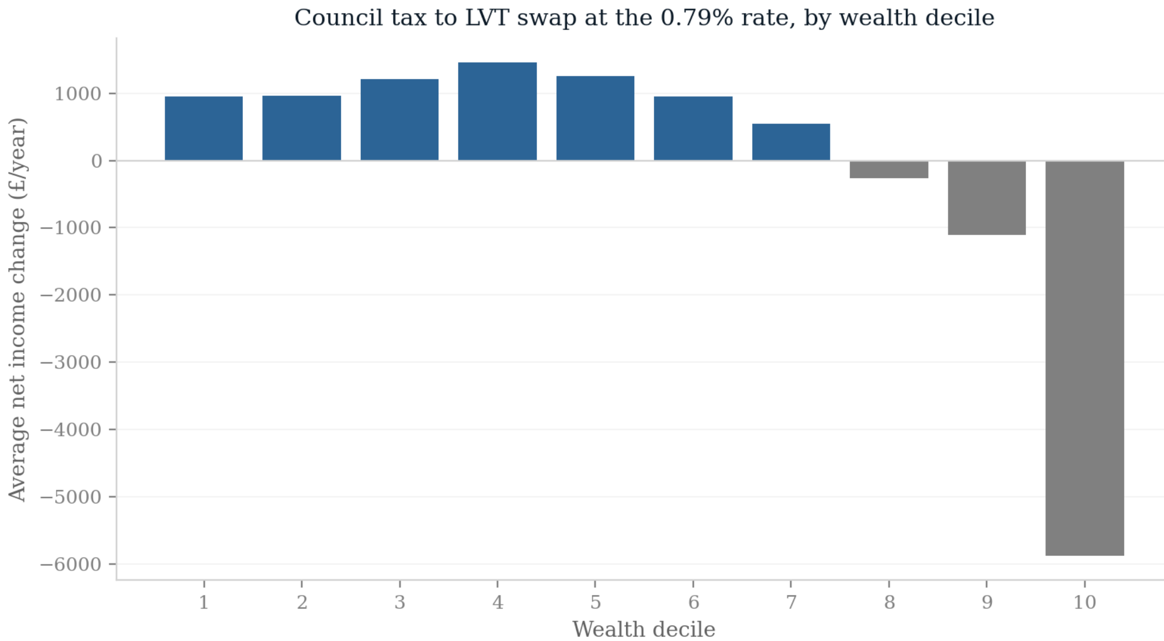

Figure 6: Average net income change by wealth decile at the 0.79% budget-neutral rate.

4.6 The joint distribution: income deciles against wealth deciles

The contrast between the two preceding tables is resolved by crossing them. Figure 7 reports the mean net income change in each cell of the income-decile × wealth-decile matrix. Within every income decile — including the top one — households in wealth deciles 1–7 gain, typically by £700–1,800 a year, and households in the top wealth decile lose, typically by £4,000–6,000 and more where high income coincides with very large land holdings. The gradient runs almost entirely along the wealth axis: reading across any row, effects flip from gains to losses between wealth deciles 7 and 9 regardless of income; reading up any column, effects barely move. The population-weighted rank correlation between household income and total wealth is 0.36, which is why a tax organised by wealth registers so weakly in statistics organised by income. This figure is the paper's central claim in a single exhibit.

4.7 Winners and losers within deciles

Decile averages conceal substantial within-decile variation, because land holdings vary widely within any given income band: an outright owner-occupier in London and a private renter in the North East may occupy the same income decile. At the budget-neutral rate, 67.4 per cent of all households gain. The share of gaining households stays between 64 and 78 per cent through deciles 1–8 before falling to just under one half in deciles 9–10 (Figure 8). Even in the top income decile, 45.4 per cent of households gain — predominantly high earners who rent or own property of modest value — while 18.2 per cent of bottom-decile households lose, typically low-income owners of valuable land. Sorted by wealth, the gradient is considerably steeper: the share of gaining households is roughly 80 per cent or more in each of the bottom seven wealth deciles and falls to 1.6 per cent in the top decile. The 17.2 per cent of households holding no land gain

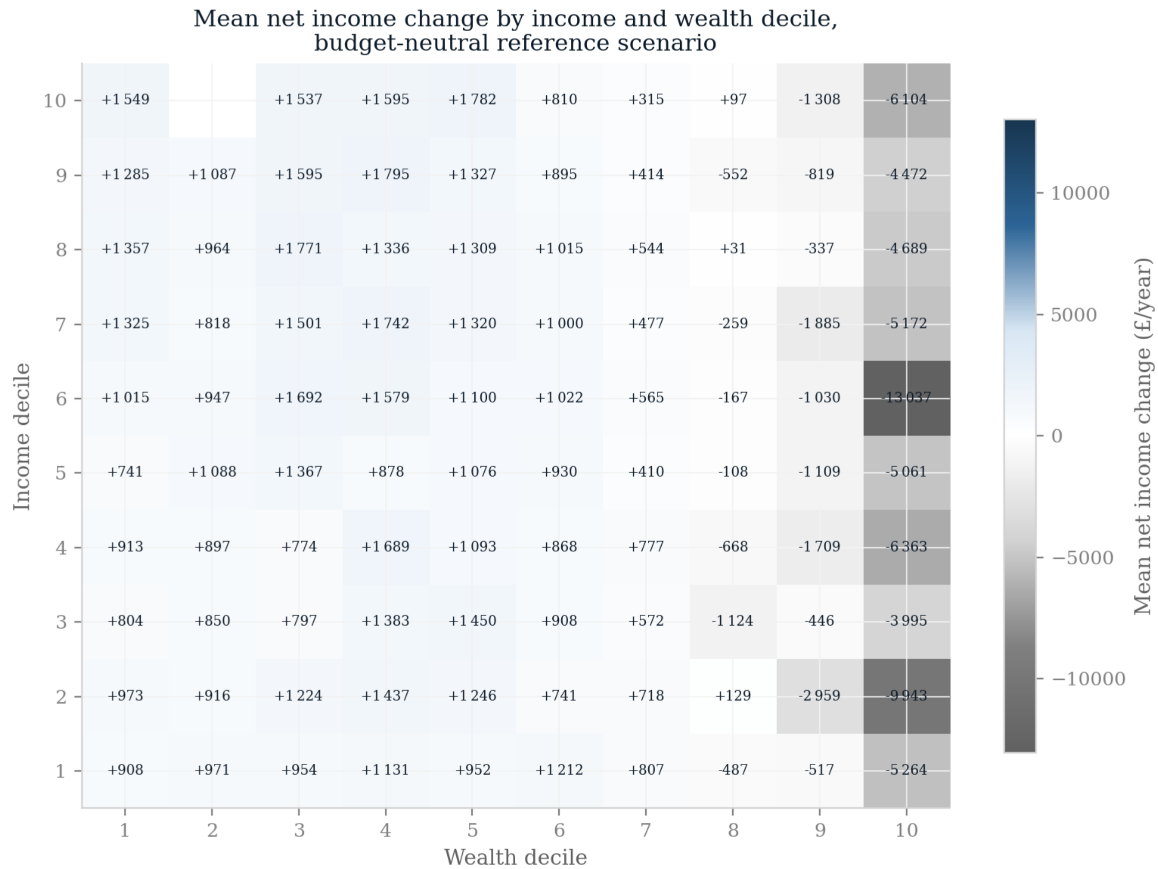

Figure 7: Mean household net income change by income decile (rows) and wealth decile (columns) at the budget-neutral 0.79 per cent rate. *Note:* income deciles are the model's equivalised-income deciles; wealth deciles are equal-weight total-wealth deciles; cells holding under 0.1 per cent of households are blank. *Source:* Authors' calculations; `analysis/submission_extras.py`.

£885 per year on average, and 77.1 per cent of them gain; the remainder are largely households whose council tax was already fully offset by council tax reduction.

The losers in the bottom of the income distribution are a recognisable group. Among households in income deciles 1–3, the 19.2 per cent that lose hold average land of £502,000 and property worth £685,000, against £52,000 and £77,000 for the 75.6 per cent that gain; 41.7 per cent contain someone of state pension age against 29.8 per cent of gainers, their oldest member averages 59.9 years against 52.7, and 43.0 per cent own outright against 33.9 per cent. They are disproportionately in London, which accounts for 18.3 per cent of these losers against 12.6 per cent of households overall. Their average loss is £2,536 a year. This is precisely the asset-rich, income-poor household of the policy debate; it is concentrated enough to be addressed by targeted transition relief, and Section 5 quantifies one such relief (an exempt band), with deferral bounded in the appendix.

That the winner share is high while the average income-decile effects are small is not a contradiction. The reform moves a fixed £58.5 billion from a base that is broad and shallow (council tax, capped at the top) to one that is narrow and deep (land, highly concentrated). Most households pay a little less; a small minority pay a great deal more. This is visible in the wealth ranking and almost invisible in the income ranking.

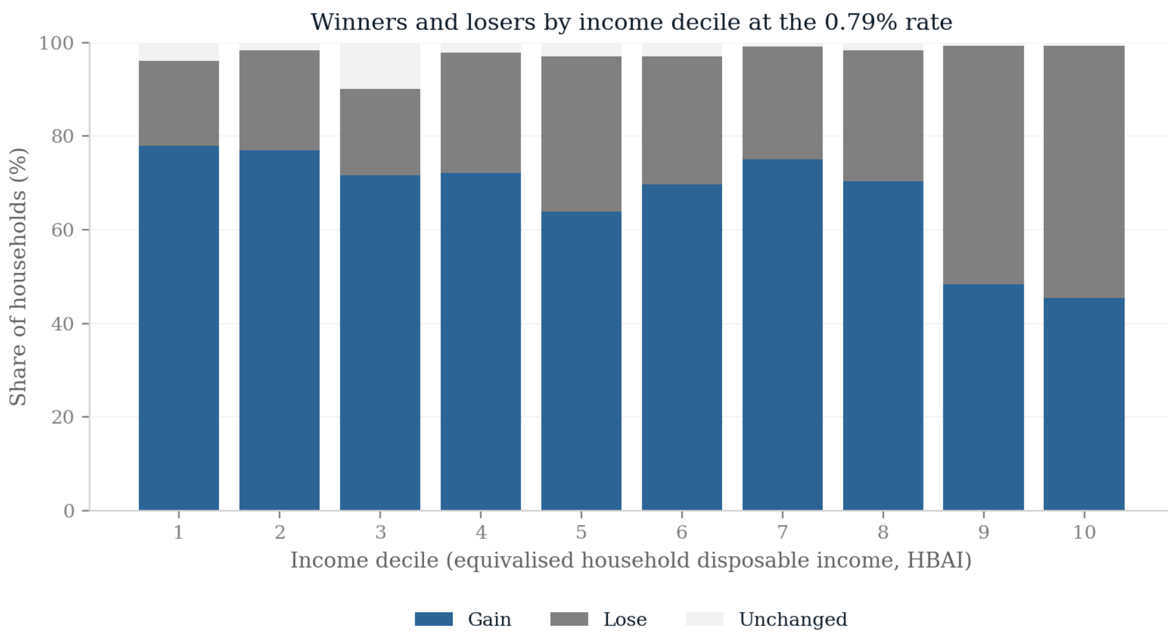

Figure 8: Share of winners and losers by income decile at the 0.79% budget-neutral rate.

4.8 Incidence by tenure

Table 8 splits the central result by tenure, which matters for two reasons: it is the margin on which the fixed-rent assumption bites, and it shows where the burden actually lands once land ownership rather than income is the organising variable.

Outright owners, 34.3 per cent of households, hold 56.6 per cent of the land and pay 56.6 per cent of the tax; they lose £755 a year on average, an aggregate of £8.3 billion. Households buying with a mortgage are almost exactly neutral (£28), holding 29.8 per cent of the land against a similar share of council tax. All three renting groups gain: private renters by £773, council tenants by £737 and housing association tenants by £235. Renters are not untaxed — they pay 13.7 per cent of the levy between them, entirely through attributed corporate land held in pensions and other financial wealth — but they retain most of an abolished council tax bill while owning almost no residential land.

The transfer is therefore, in the first instance, from outright owners to renters, with mortgagors in between. That is an intuitive description of a tax on unencumbered land ownership, and it is also where the incidence assumption does the most work. Under the pass-through mechanism that Nielsson et al. (2024) document for Denmark, some of the £8.3 billion borne by owners would be recovered from tenants in higher rents, and the £8.1 billion gain accruing to the three renting groups would shrink correspondingly. The UK has no statutory mechanism of that kind, but the magnitudes in Table 8 are what a reader should scale if they believe partial pass-through occurs.

Table 8: Impact of the 0.79% budget-neutral LVT by tenure

Tenure	Share of households (%)	Avg. land value (£)	Avg. CT saved (£)	Avg. LVT (£)	Avg. net change (£)	Winners (%)	Share of LVT (%)
Owned outright	34.3	382,634	2,253	3,008	−755	57.2	56.6
Owned with mortgage	26.7	258,520	2,060	2,032	+28	69.1	29.8
Rented privately	21.4	110,855	1,644	871	+773	76.4	10.2
Rented from council	8.8	7,904	799	62	+737	81.0	0.3
Rented from HA	8.8	82,635	884	650	+235	65.7	3.2

Note: Renters' land holdings and LVT liabilities are predominantly attributed corporate land, held through pensions and other financial wealth. *Source:* Authors' calculations; `analysis/deep_dive.py`.

4.9 Geographic incidence

Land value is the most spatially concentrated tax base in the UK, so the reform would redistribute across places at least as much as across households: land shares and land values are lower in Scotland, Wales and the North while council tax bills are not proportionately lower, and location value is concentrated in inner London and its commuter belt. Constituency-level estimates require geographic weights calibrated on the same microdata release as the rest of the paper, which are not yet available. Two calibration caveats recorded in Section 3 — the modelled Scottish and Welsh council tax overshoot, which credits those households with larger abolished bills, and the interpolated devolved-nation land shares — bear directly on any such geography. We therefore defer constituency-by-constituency results to a companion exercise on same-vintage weights rather than report them here.

4.10 Poverty and inequality

Table 9 reports poverty and inequality across the rate grid, measured against baseline-fixed absolute poverty thresholds; all rates are shares of individuals, following the HBAI convention. At the budget-neutral 0.79 per cent rate, absolute poverty after housing costs falls from 14.63 to 13.93 per cent, a reduction of 0.69 percentage points, while poverty before housing costs falls from 9.43 to 9.20 per cent (−0.23 points). The person-weighted Gini coefficient of equivalised household net income falls very slightly, by 0.0025 from a baseline of 0.2972.

The larger AHC reduction is informative rather than a puzzle. The AHC measure nets housing costs out of income, and the households lifted above the AHC line are disproportionately low-income renters, who gain their entire council tax bill and owe almost no land tax. The BHC measure does not, so it gives more weight to low-income owner-occupiers, who lose. The revenue-neutral swap therefore reduces poverty on both measures in this release, but more strongly after housing costs.

The grid shows how sensitive this is to the rate, and therefore to the replacement target. At 0.5 per cent — which is not revenue-neutral, reducing net revenue by £21.3 billion — poverty falls on both measures (BHC −0.63, AHC −0.91 points) and the Gini declines slightly (−0.0047). At 1 per cent, which raises £15.9 billion of net revenue, BHC poverty rises 0.78 points while AHC poverty is still 0.53 points below baseline. At higher rates poverty and measured income inequality rise substantially — at 5 per cent, both poverty measures rise by about 10.3 points and the Gini by 0.09 — because the static analysis levies cash charges on asset-rich, income-poor households with no offsetting use of the additional revenue. These upper-grid rows are the gross incidence of the tax alone and not complete reform packages: any practical proposal at those rates would recycle the net revenue, and the rows should not be read as forecasts of what such a package would do. The wealth Gini (0.7038) is mechanically unchanged, since the simulation does not model asset-price responses; Section 4.12 reports what capitalisation would imply for it.

Table 9: Poverty and inequality by LVT rate (council tax abolished throughout)

Rate	Poverty BHC (%)	Δ BHC (pp)	Poverty AHC (%)	Δ AHC (pp)	Δ Gini
Baseline	9.43	—	14.63	—	—
0.5%	8.80	−0.63	13.72	−0.91	−0.0047
0.79%	9.20	−0.23	13.93	−0.69	−0.0025
1.0%	10.21	+0.78	14.09	−0.53	−0.0004
1.5%	11.18	+1.75	15.15	+0.52	+0.0054
2.0%	12.41	+2.98	16.57	+1.94	+0.0129
3.0%	15.57	+6.14	19.65	+5.03	+0.0334
5.0%	19.69	+10.26	24.90	+10.27	+0.0945

Note: Absolute poverty against baseline-fixed thresholds, before (BHC) and after (AHC) housing costs, as shares of individuals (HBAI convention). Gini refers to equivalised household net income, person-weighted; baseline value 0.2972. Rates above 0.79% raise net revenue that is not recycled in the simulation. *Source:* Authors' calculations using PolicyEngine UK 2.89.4 and Enhanced FRS 2023–24 release 1.56.14.

4.11 Progressivity against income and against wealth

The decile tables and the Gini result can be combined into two summary statistics that make the paper's central claim precise. Table 10 reports Kakwani and Suits indices for council tax and for the budget-neutral LVT, each computed twice: once ranking households by income, once by wealth. The Kakwani index is the concentration index of the tax minus the Gini coefficient of the ranking variable, and is positive when the tax is progressive with respect to that variable; the Suits index is the analogous statistic computed against cumulative shares.

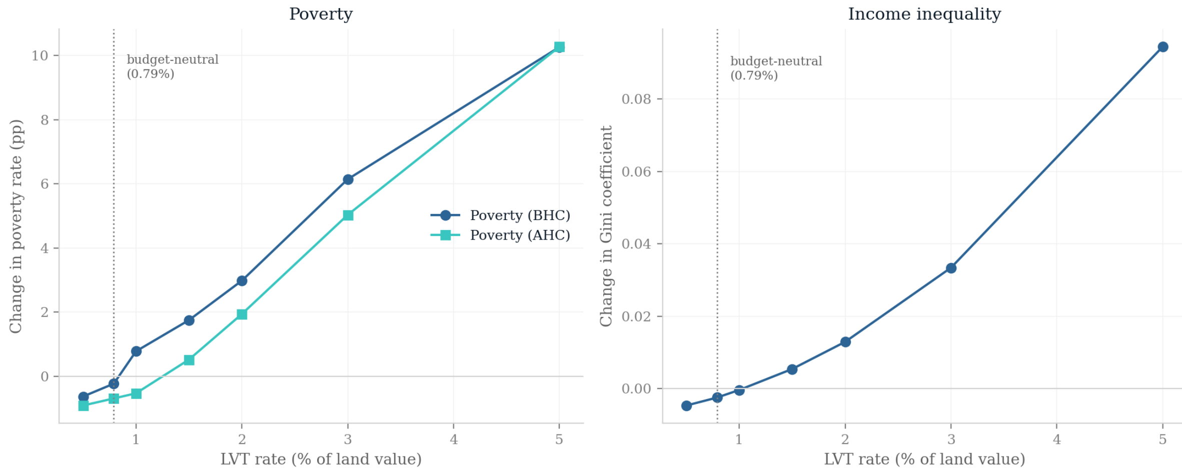

Figure 9: Poverty and Gini changes across the LVT rate grid. *Note:* council tax abolished throughout; rates above the budget-neutral 0.79 per cent raise net revenue that is not recycled, so the upper grid shows the gross incidence of the levy rather than a complete reform package. *Source:* Authors' calculations using PolicyEngine UK 2.89.4 and Enhanced FRS 2023–24 release 1.56.14; `analysis/figures.py`.

Against the *income* ranking — baseline equivalised net income after council tax, person-weighted, matching the Gini convention above, so a post-council-tax diagnostic rather than a conventional pre-tax Kakwani — both taxes are regressive: Kakwani is −0.201 for council tax and −0.084 for the LVT (Suits, −0.205 and −0.086). A land tax is regressive in income for the same reason council tax is — land, like housing, is held disproportionately by households whose current income is low relative to their assets — though the swap roughly halves the shortfall from proportionality on this axis. The improvement is nonetheless modest in level terms, which is the same conclusion the near-zero Gini change invites.

The corresponding Reynolds–Smolensky indices — the reduction in the income Gini achieved by each tax, combining its progressivity with its size — tell the same story on the income axis: −0.0072 for council tax and −0.0047 for the LVT (both taxes *raise* income inequality, the LVT by about a third less), and the swap itself lowers the Gini by 0.0025, matching the reform effect reported in Section 4.10.

Against the *wealth* ranking the two taxes are almost unrecognisably different. Council tax has a Kakwani index of −0.523: it is sharply regressive in wealth, taking a much smaller share from wealthy households than their share of wealth. The LVT has a Kakwani index of −0.032 — within rounding of proportionality. Its concentration index against wealth is 0.672, against a wealth Gini of 0.704, so the tax is distributed across households almost exactly as wealth itself is. Council tax's concentration index is 0.180, barely a quarter as concentrated. Because a small fraction of households have negative net wealth, over which cumulative-share indices are not well defined, we recompute all the wealth-ranked indices with wealth bottom-coded at zero; they are identical to four decimal places.

This is the sharpest statement of the result. Replacing council tax with a land value tax converts a steeply wealth-regressive tax into a nearly wealth-proportional one, while doing very little that conventional income-based statistics can detect. The gap between −0.523 and −0.032

is what the reform accomplishes; the gap between −0.201 and −0.084 is what a standard distributional analysis would report.

Table 10: Progressivity of council tax and of the budget-neutral LVT

	Ranked by income		Ranked by wealth	
	Kakwani	Suits	Kakwani	Suits
Council tax (net of CTR)	−0.201	−0.205	−0.523	−0.597
LVT at 0.79%	−0.084	−0.086	−0.032	−0.050
Memo: concentration index, council tax	0.097		0.180	
Memo: concentration index, LVT	0.214		0.672	
Memo: Gini of ranking variable	0.298		0.704	

Note: Kakwani = concentration index of the tax less the Gini of the ranking variable; positive indicates progressivity. Income-ranked indices use baseline equivalised net income, measured after council tax, person-weighted; wealth-ranked indices use total household wealth, household-weighted. Wealth-ranked indices are unchanged to four decimal places when wealth is bottom-coded at zero. *Source:* Authors' calculations; `analysis/deep_dive.py`.

4.12 Validation and robustness

The swap has no benefit interactions. Running the reform through the full tax and benefit model and comparing with the arithmetic $\Delta y_i = \mathrm{CT}_i - rL_i$, we find the two agree to within £0.25 for every household at every rate tested (0.5, 1.0 and 2.0 per cent), and the share of households for which they differ by more than £1 is zero. Under the modelled implementation, the swap changes no means-tested entitlement: council tax reduction is abolished with the tax, and the LVT does not enter the income or capital tests for Universal Credit, Pension Credit or the legacy benefits — a design choice of the modelled policy, since a hypothetical levy has no legislated benefit treatment, rather than a statutory fact. The consequence is that the microdata, not the tax-benefit calculator, do all the work in this reform — the calculator's contribution is the council tax baseline, the equivalisation and the HBAI-style poverty thresholds. We regard this as worth stating explicitly because it is easy to assume otherwise, and it means the results can be reproduced and audited without re-running the model.

Statistical inference. Table 11 attaches bootstrap standard errors and 95 per cent intervals to the headline statistics, using 1,000 Poisson weight-bootstrap replicates with the budget-neutral rate re-solved on each resampled population. Under this perturbation scheme the results the paper leans on are stable: the winner share is 67.4 per cent ± 1.2, the top wealth decile's loss is bounded well away from zero, the after-housing-cost poverty interval excludes zero, and the wealth-ranked Kakwani indices of the two taxes are tightly estimated and nowhere near overlapping. Equally instructive are the two statistics that are *not* significant at the 5 per cent level: the before-housing-cost poverty change and (marginally) the Gini change. That is the paper's thesis in inferential form — on the income-based statistics that dominate UK distributional debate, the reform is indistinguishable from doing nothing under these intervals, while on the wealth axis its effects are estimated tightly. As with the decile intervals above, these capture sampling variation in the survey weights only; the land-value imputation is held fixed across replicates, so total uncertainty is understated.

Table 11: Bootstrap inference on the headline statistics, budget-neutral scenario

Statistic	Point	SE	95% interval
Budget-neutral rate (%)	0.79	0.04	[0.71, 0.87]
Winner share (%)	67.4	1.2	[64.9, 69.5]
Income decile 1 mean (£)	+584	134	[+301, +812]
Income decile 10 mean (£)	−1,429	338	[−2,059, −788]
Wealth decile 10 mean (£)	−5,871	330	[−6,530, −5,187]
Δ Poverty BHC (pp)	−0.23	0.35	[−0.80, +0.59]
Δ Poverty AHC (pp)	−0.69	0.26	[−1.22, −0.20]
Δ Gini	−0.0025	0.0013	[−0.0047, +0.0001]
Kakwani (wealth), council tax	−0.523	0.010	[−0.541, −0.504]
Kakwani (wealth), LVT	−0.032	0.007	[−0.044, −0.019]

Note: 1,000 replicates; each multiplies every household weight by an independent Poisson(1) draw and re-solves the budget-neutral rate. Percentile intervals. Sampling variation only — the land-value imputation is fixed across replicates. *Source:* Authors' calculations; `analysis/bootstrap_ci.py`.

Robustness. Table 13 varies the replacement target, the composition of the base and the instrument. The qualitative results are stable. The share of households gaining stays within a 65–70 per cent band across every specification. The budget-neutral rate ranges from 0.72 to 1.09 per cent, the top of that range being the household-land-only base, which is 28 per cent smaller. Calibrating the replacement target to OBR receipts rather than modelled liabilities lowers the rate to 0.72 per cent and, mechanically, improves every reported group's outcome relative to the central specification (the top income and wealth deciles still lose in level terms) — it is a £4.8 billion net tax cut rather than a neutral swap, and we therefore treat it as a bound rather than a competing central case. Excluding Northern Ireland, whose households would otherwise pay the land tax on top of domestic rates, raises the rate to 0.80 per cent and leaves the distribution essentially unchanged. Assuming half of corporate equity is foreign-owned and therefore outside the household base raises the required rate to 0.91 per cent — the largest calibration sensitivity apart from the household-land-only base — but still leaves 67.2 per cent of households gaining. Perturbing every regional land share by ±10 per cent, or rescaling household land to the un-uprated ONS 2024 benchmark, changes the rate by at most 0.06 percentage points and the decile effects by tens of pounds. Within-region heterogeneity in land shares, which the single λ_r suppresses, is bounded by a Monte Carlo exercise: drawing an independent mean-preserving lognormal land share for every household, at coefficients of variation of 0.1, 0.2 and 0.3 around the regional value, and re-solving the rate on each of 100 draws, moves the budget-neutral rate by 0.01 percentage points, the winner share by half a percentage point, and the top wealth decile's loss by under £100 (across-draw standard deviations) at even the widest dispersion (`analysis/lambda_mc.py`). Decile-level results are insensitive to within-region land-share heterogeneity because the noise averages out within cells; individual liabilities, of course, do not. Assigning Northern Ireland the superseded population-weighted English mean land share (0.67) rather than the central interpolation from the price-similar English regions (0.44, as for Scotland and Wales) grows the base by only £0.03 trillion — Northern Ireland is 2.4 per cent of households — and leaves every summary statistic essentially unchanged, although it raises Northern Irish liabilities by roughly a half. Excluding directly owned land (A_i in equation 2) — the component flagged in Section 3.3 as unstable across data releases — removes £0.67 trillion from the base, raises the rate to 0.86 per cent, and leaves 65.4 per cent of households gaining; the qualitative results survive the most conservative treatment of that component. Finally, because the two calibration gaps discussed in Section 3 — the council tax overshoot relative to OBR receipts and the household-land overshoot relative to the un-uprated ONS benchmark — push the required rate in the same direction, we close both at once: targeting OBR receipts on the ONS-scaled base gives a rate of 0.76 per cent and results between the two single-adjustment rows.

The final row of Table 13 is a comparator rather than a robustness check: a proportional tax on total property value, structures included, at the 0.74 per cent rate that raises the same revenue. It is less progressive on the wealth ranking — the top wealth decile loses £4,310 rather than £5,871 — and it produces fewer winners (63.4 against 67.4 per cent), because the base is spread more evenly across the property-owning population instead of being concentrated where location value is highest. That difference is the quantitative content of the land/structures distinction.

Incidence variants. Two further variants relax the incidence assumptions rather than the calibration. First, the central specification charges the corporate component of the levy (£16.2 billion at the budget-neutral rate) to households as a same-year cash reduction in disposable income, in proportion to their corporate wealth — including wealth held through pensions, where the first-round effect would in practice fall on the (often inaccessible) pension asset rather than on current cash income. Treating the corporate component instead as a stock levy on corporate wealth, with only the household-land component entering current income, raises the winner share to 75.8 per cent and cuts the top wealth decile's income-side loss to £3,224; the corporate charge is then analogous to the capitalisation effects discussed below, amounting to 0.06 per cent of total household wealth per year. Second, we treat rent pass-through as a sensitivity dimension rather than a single variant. Table 12 reports incidence as a function of the share θ of the levy on privately rented dwellings shifted to tenants, from the fixed-rent central case ($\theta = 0$) to full pass-through ($\theta = 1$), an upper bound motivated by the close-to-full (around 83 per cent) pass-through Löffler and Siegloch (2024) estimate for recurrent property taxes in German data, using the regional mean owner-occupier land value as a proxy for the (unobserved) land under each rented dwelling and rebating the same aggregate to owners so total revenue is unchanged. The winner share is robust — it falls only from 67.4 to 63.4 per cent across the whole range — but the after-housing-cost poverty reduction is not: it shrinks from 0.69 percentage points at $\theta = 0$ to 0.23 at $\theta = 0.5$ and reverses sign between $\theta = 0.5$ and $\theta = 0.75$, as low-income private renters progressively hand back their council tax savings in higher rents. The poverty result, much more than the winner share, is what the fixed-rent assumption protects, and readers who find the Danish and German pass-through evidence persuasive should discount it accordingly.

Capitalisation. The central results are flow accounting and assume no capitalisation. If instead the levy is fully anticipated and permanent, it capitalises into land prices in the manner first documented empirically by Oates (1969): a rate r discounted at ρ implies, for a levy assessed on the fixed pre-reform base, a proportional fall of r/ρ in land values; a levy reassessed on post-

Table 12: Incidence as a function of the rent pass-through share θ

θ	Shifted to tenants (£bn)	Winners (%)	Private renter mean (£)	Δ Poverty BHC (pp)	Δ Poverty AHC (pp)
0.00	0.0	67.4	+773	−0.23	−0.69
0.25	4.0	66.7	+193	−0.32	−0.39
0.50	8.0	66.8	−386	−0.34	−0.23
0.75	11.9	64.9	−966	−0.29	+0.24
1.00	15.9	63.4	−1,545	−0.02	+0.43

Note: Budget-neutral 0.79 per cent rate throughout; the amount shifted to tenants is rebated to owner households in proportion to household land, so aggregate revenue is unchanged. "Private renter mean" is the mean net income change of private-renter households. *Source:* Authors' calculations; `analysis/submission_extras.py`.

Table 13: Robustness: revenue-neutral rate and distributional summary under alternative assumptions

Specification	Base (£tn)	Rate (%)	Winners (%)	Decile 1 (£)	Decile 10 (£)	Wealth 10 (£)	Δ Poverty AHC (pp)
Central	7.44	0.79	67.4	+584	−1,429	−5,871	−0.69
OBR receipts target	7.44	0.72	68.6	+656	−1,111	−5,178	−0.79
Great Britain only	7.34	0.80	66.7	+575	−1,477	−5,861	−0.67
Household land only	5.38	1.09	68.6	+529	−1,225	−5,478	−0.69
Half of corporate land foreign-owned	6.41	0.91	67.2	+561	−1,343	−5,706	−0.69
Household land scaled to ONS 2024	7.08	0.83	67.4	+587	−1,439	−5,891	−0.75
Land shares −10%	6.90	0.85	67.3	+588	−1,445	−5,902	−0.75
Land shares +10%	7.97	0.73	67.3	+580	−1,415	−5,845	−0.69
NI legacy land share (0.67)	7.47	0.78	67.4	+587	−1,413	−5,874	−0.69
Excluding directly owned land	6.77	0.86	65.4	+603	−1,695	−5,236	−0.80
OBR target on ONS-scaled base	7.08	0.76	68.6	+658	−1,121	−5,196	−0.78
Comparator: proportional property tax	7.91	0.74	63.4	+554	−1,069	−4,310	−0.77

Note: Each row sets the rate so that revenue equals the replacement target on that row's base. "Decile 1" and "Decile 10" are average net income changes by income decile; "Wealth 10" is the top wealth decile. *Source:* Authors' calculations using PolicyEngine UK; `analysis/extensions.py`.

capitalisation values falls by $r/(\rho + r)$ instead, so these figures are upper bounds. At $\rho = 5$ per cent the 0.79 per cent rate implies a 15.7 per cent fall in land prices and an aggregate wealth loss of £1.17 trillion; at $\rho = 3$ per cent, a 26.2 per cent fall and £1.95 trillion. This is a transfer to the state of an asset value that was itself partly the capitalisation of past public investment, and under the textbook assumption it lands overwhelmingly on existing owners at announcement rather than on subsequent purchasers. Whether it does so in practice is contested: Høj et al. (2018) find full capitalisation in Danish data, whereas Nielsson et al. (2024) cannot reject zero and argue the burden is shared with tenants and future buyers. We report the arithmetic under full capitalisation because it bounds the wealth effect, not because the evidence compels it. Table 18 in the appendix reports the implied one-off wealth loss by wealth decile: at $\rho = 3$ per cent the top wealth decile loses £284,431 of land value, 6.6 per cent of its total wealth, against £329 for decile 3. Two points follow. First, the wealth-decile gradient of the reform is steeper still on a stock than on a flow basis. Second, the flow results in Sections 4.4–4.5 should be read as impact incidence in the first years, before the announcement effect on prices works through.

5 Reform packages and comparators

The budget-neutral reference scenario is a bare swap: council tax out, land value tax in, nothing else. Real proposals are rarely bare, and the framework is not restricted to revenue neutrality. This section simulates four scenarios beyond the reference point — recycling the proceeds of a higher rate, exempting a band of land value, broadening the set of taxes replaced, and the instrument the government has actually chosen — on the same microdata and with the same arithmetic.

5.1 Recycling: a land value tax with a citizen's dividend

The rate grid in Section 4.10 levies rates above 0.79 per cent without spending the proceeds, which is why poverty rises so steeply across it. That is not a proposal anyone has made. George (1879) argued for taxing land in order to fund public provision and to displace other taxes, and the modern Georgist formulation returns the surplus as a universal payment. Table 14 therefore repeats the grid with the net revenue above the council tax replacement returned as an equal per-person dividend.

The change is substantial. At 2 per cent, the unrecycled levy raised before-housing-cost poverty by 2.98 percentage points; recycled, it *reduces* it by 0.80 points and reduces after-housing-cost poverty by 2.79 points, while paying a dividend of £1,266 per person and leaving 70.4 per cent of households better off. At 1.5 per cent the package reduces poverty on both measures (BHC −0.75, AHC −2.22) and gives the poorest income decile £1,181 a year. The reason is mechanical but worth stating: the levy is drawn from a base concentrated at the top of the wealth distribution and returned on a per-head basis, so the net transfer runs down the distribution even though the tax itself is only proportional to wealth.

Two qualifications. First, the dividend is a flat cash payment and we do not model any interaction with means-tested benefits, so these figures overstate the gain for households whose entitlements would taper. Second, the poverty gains do not grow monotonically with the rate: the before-housing-cost gain peaks around 2 per cent and shrinks at 3 and 5 per cent (BHC −0.42 and −0.20), and measured income inequality turns up at 5 per cent (Gini +0.014), because the liabilities falling on asset-rich, income-poor households increasingly offset a per-head payment.

The recycled package improves on the unrecycled one at every rate, but the improvement is not unbounded in the rate.

Table 14: LVT with net revenue returned as an equal per-person dividend

Rate (%)	Recycled (£bn)	Dividend (£/person)	Winners (%)	Δ Poverty BHC (pp)	Δ Poverty AHC (pp)	Decile 1 (£)
0.79	0.0	0	67.4	−0.23	−0.69	+584
1.0	15.9	223	71.9	−0.03	−1.35	+763
1.5	53.1	745	71.1	−0.75	−2.22	+1,181
2.0	90.3	1,266	70.4	−0.80	−2.79	+1,599
3.0	164.6	2,309	69.4	−0.42	−3.63	+2,434
5.0	313.4	4,395	68.4	−0.20	−3.03	+4,106

Note: Council tax is abolished in every row. Revenue above the £58.5 billion replacement is divided equally across all individuals. No benefit interactions are modelled on the dividend. *Source:* Authors' calculations; `analysis/deep_dive.py`.

5.2 An exempt band of land value

Section 6 notes an exempt band as a standard mitigation for the liquidity problem. Table 15 costs it. Exempting the first £100,000 of land value per household removes 51.7 per cent of households from the tax altogether and shrinks the base from £7.44 to £5.55 trillion, so revenue neutrality requires 1.05 rather than 0.79 per cent. The distributional effect of the exemption is to sharpen the reform, not to soften it: winners rise from 67.4 to 69.4 per cent, the poorest decile's gain rises from £584 to £690, and the top wealth decile's loss rises from £5,871 to £7,727.

This is worth emphasising because the intuition often runs the other way. An exempt band is usually discussed as a concession; here it is a progressivity instrument, because it removes small landholders from the base and concentrates the same revenue on large ones. At a £200,000 band, 64.5 per cent of households pay nothing, nearly three-quarters gain, and the top wealth decile loses £9,584. The cost is a higher headline rate and a narrower, more concentrated base — which is also a more volatile one, and more exposed to valuation dispute at the top.

Table 15: Revenue-neutral LVT with a per-household exempt band of land value

Exempt band (£)	Taxable base (£tn)	Rate (%)	Households untaxed (%)	Winners (%)	Decile 1 (£)	Wealth 10 (£)
0	7.44	0.79	17.2	67.4	+584	−5,871
50,000	6.41	0.91	40.6	68.4	+639	−6,786
100,000	5.55	1.05	51.7	69.4	+690	−7,727
200,000	4.23	1.38	64.5	73.5	+754	−9,584

Note: Each row replaces the same £58.5 billion. "Untaxed" includes the 17.2 per cent of households holding no land. *Source:* Authors' calculations; `analysis/deep_dive.py`.

5.3 Broadening the taxes replaced

Mirrlees et al. (2011), Muellbauer (2005) and Leunig (2024) all propose replacing stamp duty alongside council tax, on the grounds reviewed in Section 2: a transaction tax distorts mobility and the timing and volume of sales (Best and Kleven, 2018; Hilber and Lyytikäinen, 2017), and none of that distortion has any counterpart under a recurrent charge on an immobile base. Table 16 costs the broader packages.

Adding stamp duty, which the model puts at £11.4 billion, raises the replacement target to £69.9 billion and the revenue-neutral rate to 0.94 per cent. Fewer households gain (63.9 against 67.4 per cent), which is unsurprising: stamp duty is paid by a small number of movers in any given year, so replacing it with a recurrent charge spreads a concentrated liability across all landholders. The efficiency argument for doing so is strong and the distributional argument is weaker — a trade-off the proposals cited above generally acknowledge. Folding in Northern Irish domestic rates (£0.6 billion) leaves the rate essentially unchanged at 0.79 per cent, and slightly improves the distribution by removing the double taxation of Northern Irish households that our central specification imposes.

Table 16: Alternative replacement targets

Taxes replaced	Revenue (£bn)	Rate (%)	Winners (%)	Decile 1 (£)	Decile 10 (£)	Wealth 10 (£)
Council tax (central)	58.5	0.79	67.4	+584	−1,429	−5,871
Council tax and NI domestic rates	59.0	0.79	68.8	+598	−1,461	−5,942
Council tax and stamp duty	69.9	0.94	63.9	+475	−1,218	−5,263
All three	70.4	0.95	65.5	+489	−1,249	−5,334

Note: Stamp duty is modelled as an annual flow of liabilities on the transactions occurring in the simulation year, and replacing it with a recurrent charge therefore redistributes from movers to holders in a way this static comparison captures only in the first year. *Source:* Authors' calculations; `analysis/deep_dive.py`.

5.4 The instrument the government chose

The High Value Council Tax Surcharge (HM Treasury, 2025) addresses the same defect as our reform — that council tax stops discriminating above Band H — with a very different instrument. Simulating its announced parameters on our microdata gives a useful check and a sharp comparison. We find 171,000 liable properties raising £0.56 billion, against the OBR's 165,000 and £0.4 billion; the closeness of the property count is reassuring for the upper tail of our wealth data, and our slightly higher revenue is consistent with the OBR's costing being net of behavioural response and non-compliance, which we do not model.

The comparison with the LVT is stark on every dimension. The surcharge touches 0.53 per cent of households and raises 1.0 per cent of what the land tax raises. All of it falls on the top wealth decile, but as an average across that decile it amounts to £172 a year, against £8,533 under the LVT. It reaches only 7.4 per cent of its revenue from the top *income* decile, confirming that high-value property and high current income are different things. And because it is a flat charge within wide bands, it reintroduces at £2 million exactly the cliff-edge and value-insensitivity that make the existing band structure regressive: a £2.6 million and a £3.4 million property both pay £3,500.

None of this makes the surcharge a bad measure — it raises revenue from an under-taxed group at low administrative cost, and it establishes that the Valuation Office can value high-value homes at current prices, which is the precondition for anything more ambitious. But it is a patch of roughly one per cent the size of the reform considered here, and it leaves the structure it patches intact.

6 Discussion

6.1 Summary of findings

In the budget-neutral reference scenario, replacing council tax with a 0.79 per cent flat land value tax leaves roughly two thirds of households better off, reduces absolute poverty after housing costs by 0.69 percentage points, and shifts the tax burden substantially up the wealth distribution, while leaving the Gini coefficient of equivalised household net income essentially unchanged. The dissociation between the wealth-decile and income-Gini results is the paper's main analytical point, and it holds across the scenario space: distributional statistics computed over the income distribution can be nearly uninformative about a reform whose incidence is organised by wealth. An LVT is a tax on a stock that is highly concentrated — we estimate that the top wealth decile holds 47 per cent of all land, and the wealth Gini is 0.70 against 0.30 for equivalised income — assessed on households whose position in the income distribution is only weakly related to that stock. We are correspondingly cautious about the income-decile profile: average effects in deciles 2–8 are a few hundred pounds a year, not monotonic in income, and rest on imputed land values whose decile means are visibly noisy; the data support the claim that the reform is roughly neutral across the bottom eight income deciles and costly to the top two, not claims about particular middle deciles. The efficiency case, which our static framework does not quantify, points the same way: unlike council tax, an LVT imposes no charge on constructing, extending or upgrading property, and its base is immobile (George, 1879; Oates and Schwab, 1997).

6.2 Limitations

Several caveats qualify the results.

Static incidence. We assign the full burden to current landowners, with no behavioural response and no capitalisation in the central results. Standard theory predicts that an anticipated recurrent land tax capitalises into lower land prices (Feldstein, 1977); the empirical literature does not settle the extent, with Høj et al. (2018) finding full capitalisation in Danish data and Nielsson et al. (2024) a precise zero on later variation. Section 4.12 quantifies the textbook upper bound — full capitalisation at 3–5 per cent discount rates implies a fall in land values of 15–26 per cent, concentrated on the top wealth decile — and reports incidence as a function of the rent pass-through share: the winner share falls only from 67.4 to 63.4 per cent between zero and full pass-through, but the after-housing-cost poverty reduction shrinks from 0.69 to 0.23 percentage points at half pass-through and reverses sign above it, a margin on which the German evidence of Löffler and Siegloch (2024) points towards substantial pass-through. The poverty result therefore depends on the fixed-rent assumption in a way the winner share does not.

Regional land shares. Equation (2) applies a single land share λ_r to every property in a region, so we understate within-region heterogeneity in individual LVT liabilities. The Monte

Carlo exercise in Section 4.12 shows that decile-level averages, the winner share and the budget-neutral rate are essentially invariant to this — household-level dispersion up to a coefficient of variation of 0.3 moves them within their bootstrap intervals — so the limitation bears on statements about individual households, not on the distributional results.

Corporate land allocation. We allocate £2.06 trillion of corporate land to households in proportion to their measured corporate wealth, abstracting from foreign ownership of UK equity (which would shift part of the burden abroad) and treating the charge as a same-year reduction in disposable income although much of the wealth is held through pensions. Section 4.12 shows that treating the corporate component as a stock levy instead raises the winner share, so the central results are conservative on this margin.

Local government finance. Council tax funds local authorities within an equalisation system (Amin-Smith and Phillips, 2019); our simulation replaces its revenue at the national level and is silent on assignment. A locally levied LVT would tie councils to an extremely uneven base — average household land value in London is 2.4 times that in the North East — and would require a substantially more aggressive equalisation regime (Scottish Government, 2015). This first-order design question is out of scope rather than resolved.

Northern Ireland. The reference scenario leaves domestic rates unchanged, so Northern Irish households pay LVT in addition to their existing property tax; Section 5.3 folds rates into the replacement and Section 4.12 reports the Great Britain-only alternative.

Measurement. Land values are imputed rather than observed: property wealth is self-reported, the survey's upper tail is thin, and the National Balance Sheet's land estimates rest on a residual valuation method (ONS, 2025). Council tax liabilities are modelled from regional band averages, and the modelled Scottish and Welsh totals run roughly a fifth above devolved budget figures, which inflates the estimated gains in those nations (Section 4.9). Two further cautions apply. In June 2025 the Office for Statistics Regulation removed the accredited official statistics status of Wealth and Assets Survey outputs on quality grounds (OSR, 2025), citing declining response rates and under-investment; since the entire wealth side of our imputation rests on the WAS, this designation change bears on every land value in the base. And the ONS's preliminary balance-sheet estimates for 2025 show household land values flat to falling (ONS, 2026), so the ONS-scaled robustness variant in Section 4.12 may be closer to current conditions than the uprated central base.

6.3 Policy implications and design options

The most salient losing group comprises asset-rich, income-poor households — typically older owner-occupiers of valuable homes, concentrated in London and the South East. Established remedies address this liquidity problem, and we quantify two of them. An exempt band of land value (Section 5.2) makes the reform more progressive rather than less: exempting the first £100,000 removes half of all households from the tax and increases both the poorest decile's gain and the top wealth decile's loss. Deferral until sale or death, as recommended by Mirrlees et al. (2011) and Muellbauer (2018), is bounded in the appendix: pensioner-household liabilities are roughly a third of first-year receipts, an upper bound on the transitional cash-flow cost rather than a revenue loss. The underlying difficulty is one of liquidity rather than solvency, since the households concerned are by construction wealthy.

Two comparators locate the LVT within the wider reform menu. Broadening the replacement to include stamp duty (Section 5.3), as Mirrlees et al. (2011), Muellbauer (2005) and Leunig (2024) propose, raises the neutral rate and lowers the winner share, since a liability concentrated on movers is spread across all landholders. A proportional tax on total property value (Muellbauer, 2005; Corlett and Gardiner, 2018), simulated in Section 4.12, shares the LVT's correction of council tax's regressivity and is administratively simpler, but produces fewer winners and a shallower wealth gradient: location value is more concentrated than structure value, so for a given revenue target the LVT asks more of the largest landholders and less of everyone else.

6.4 Conclusion

This paper has presented, to our knowledge, the first open-source household-level microsimulation framework for land value tax reform in the United Kingdom, applied to a family of scenarios for replacing council tax. In the budget-neutral reference scenario at 0.79 per cent, roughly two thirds of households gain, after-housing-cost poverty falls modestly, and the burden of recurrent property taxation moves from council tax's regressive banded structure onto the distribution of land wealth, while measured income inequality is essentially unaffected — a caution against reading near-zero movement in headline income statistics as evidence that little has changed. Measured against wealth, council tax is sharply regressive (Kakwani −0.523) while the land value tax is close to proportional (−0.032); measured against income both remain regressive and the gap is far smaller, so the choice of ranking variable, not the reform, drives most of what a conventional distributional table reports. Across scenarios, the design choices usually discussed as concessions are not: an exempt band makes the reform more progressive, and recycling the proceeds of a higher rate as a per-person dividend reduces poverty on both measures. The principal obstacles to implementation are practical rather than analytical — valuing land separately from structures at scale, and managing the transition for asset-rich, income-poor owners — and both admit established policy responses, two of which we cost. Extending the framework to incorporate capitalisation, behavioural responses and parcel-level land valuation remains a subject for future research.

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A Appendix

A.1 Impact by income decile at alternative rates

Table 17 reports the average net income change by income decile across the full rate grid, with council tax abolished in every scenario. At 0.5 per cent every decile except the top gains on average, and the top is essentially unchanged (−£7), the reform reducing net revenue by £21.3 billion; from 1.5 per cent upwards every decile loses on average, with losses concentrated in deciles 9–10. Rates other than 0.79 per cent are not revenue-neutral, so these columns combine the incidence of the swap with that of a net tax cut or rise.

Table 17: Average net income change (£/year) by income decile and LVT rate

Decile	0.5%	0.79%	1.0%	1.5%	2.0%	3.0%	5.0%
1	+906	+584	+344	−218	−780	−1,903	−4,150
2	+568	−17	−454	−1,477	−2,500	−4,545	−8,636
3	+929	+640	+425	−80	−584	−1,593	−3,611
4	+639	+122	−264	−1,168	−2,071	−3,877	−7,490
5	+562	−87	−573	−1,707	−2,841	−5,110	−9,647
6	+789	+173	−288	−1,365	−2,442	−4,595	−8,903
7	+931	+385	−24	−978	−1,933	−3,842	−7,659
8	+894	+161	−387	−1,668	−2,950	−5,512	−10,637
9	+387	−668	−1,458	−3,302	−5,146	−8,835	−16,212
10	−7	−1,429	−2,493	−4,979	−7,465	−12,437	−22,382

Note: Rates above 0.79% raise net revenue that is not recycled; rates below it reduce net revenue. *Source:* Authors' calculations using PolicyEngine UK.

A.2 Landless households

Households holding no land — no residential property, directly owned land or corporate wealth — make up 17.2 per cent of households. Because their LVT liability is zero at any rate, their outcome is invariant across the rate grid: an average gain of £885 per year (their abolished council tax bill), with 77.1 per cent winning. The non-winners are predominantly households whose council tax was already fully covered by council tax reduction, for whom abolition changes nothing.

A.3 Worked single-household example

Consider a household owning a £400,000 home in London, where the land share is 85 per cent, implying household land of £340,000. If the household also holds £50,000 of corporate wealth, its attributed corporate land is that sum multiplied by the population ratio of aggregate corporate land to aggregate corporate wealth, which the simulation puts at £0.14514 per pound of corporate wealth, or £7,257. Its total land value is therefore £347,257 and its annual liability at the budget-neutral rate (0.786 per cent before rounding) is £2,730; at a 1 per cent rate it would be £3,473. Whether the household gains depends on the abolished council tax bill: with an illustrative upper-band London bill of £2,400 it loses roughly £330 a year, and it breaks even at a bill of £2,730.

The allocation ratio must be computed against the weighted population. Evaluating equation (2) inside a single-household simulation assigns the entire £2.06 trillion corporate land base to that one record, because it constitutes the whole population; the example above therefore uses the population ratio from the microsimulation.

A.4 Capitalisation by wealth decile

Table 18 reports the one-off wealth loss implied by full capitalisation of the permanent 0.79 per cent levy, $\Delta W_i = -(r^*/\rho)L_i$, at discount rates of 3, 4 and 5 per cent (the table reports the 3 and 5 per cent columns), under the assumption that the levy is assessed for ever on the fixed pre-reform base — a levy reassessed on post-capitalisation values falls by $r^*/(\rho + r^*)$ instead, so these are upper bounds. The aggregate loss is £1.17 trillion at $\rho = 5$ per cent, £1.46 trillion at 4 per cent and £1.95 trillion at 3 per cent, corresponding to falls in land prices of 15.7, 19.7 and 26.2 per cent. Expressed as a share of total wealth the loss peaks in the middle of the wealth distribution — deciles 5 and 6, whose wealth is overwhelmingly housing — while in absolute terms it is concentrated at the top.

Table 18: Capitalised one-off wealth loss by wealth decile (£), 0.79% rate

Wealth	$\rho = 3\%$		$\rho = 5\%$	
decile	Loss (£)	% of wealth	Loss (£)	% of wealth
1	0	—	0	—
2	8	2.9	5	1.7
3	329	3.7	197	2.2
4	6,574	8.1	3,944	4.9
5	20,017	9.8	12,010	5.9
6	36,301	9.5	21,780	5.7
7	55,369	8.4	33,222	5.1
8	83,911	8.0	50,347	4.8
9	117,776	6.9	70,666	4.1
10	284,431	6.6	170,659	4.0

Note: Loss is $(r^*/\rho)L_i$ averaged within decile; "% of wealth" is that loss over average total wealth. Deciles 1–2 hold negligible land. *Source:* Authors' calculations; `analysis/extensions.py`.

A.5 Deferral for pensioner households

The liquidity problem the reform creates is concentrated and measurable. Households containing at least one person of state pension age make up 27.2 per cent of households and would owe £20.6 billion of the £58.5 billion total, or 35.2 per cent of the yield, at an average liability of £2,354. This is an upper bound on what a deferral scheme could attract: it counts every pensioner household's full liability, including attributed corporate land and households that would not elect to defer, and models no eligibility test, take-up, interest or settlement flows. A full deferral option — the liability rolled up with interest against the property, as recommended by Mirrlees et al. (2011) — would therefore postpone at most roughly a third of receipts in the first year, declining thereafter as deferred estates are settled. This bounds the transitional financing problem: it is a cash-flow cost to government rather than a revenue loss.

A.6 Data and parameter summary

Table 19: Key data sources and parameters

Item	Value	Source / method
Simulation		
Microdata	EFRS 2023–24	FRS enhanced with SPI quantile-forest imputation and WAS wealth
Model version	policyengine-uk 2.89.4	Released version used; see replication note below
Simulation year	2026–27	Uprated from the 2023–24 survey
Households	32.08m	Weighted EFRS household count
Council tax baseline (replacement target)		
Gross liabilities	£61.9bn	Modelled, VOA band calibration
Council tax reduction	£3.4bn	Modelled
Net revenue (R_{CT})	£58.5bn	Equation (1)
Memo: OBR receipts	£53.7bn	OBR EFO; robustness row
Land base		
Household land	£5.38tn	Model output, equation (2)
Corporate land	£2.06tn	ONS NBS aggregate, allocated by corporate wealth
Total base (L)	£7.44tn	104.8% of un-uprated ONS 2024 (£7.10tn)
Regional land shares (λ_r)	0.42–0.85	MHCLG residual method, equation (3)
Reform		
Budget-neutral rate (r^*)	0.79%	R_{CT}/L
LVT parameter	`gov.contrib.ubi_center.land_value_tax.rate`	
Abolition parameter	`gov.contrib.abolish_council_tax`	

A.7 Replication note

All results are generated from a single baseline simulation of PolicyEngine UK 2.89.4 on the pinned Enhanced FRS release `enhanced_frs_2023_24@1.56.14`, together with the closed-form arithmetic validated in Section 4.12. The generating script (`uk_lvt/pipeline_direct.py`) reruns the full model at three rates on every invocation and refuses to write results if the arithmetic and the model disagree by more than £1 per household or 0.01 percentage points of poverty.

Three caveats on measurement and version sensitivity are worth recording. First, all poverty rates are shares of individuals (household poverty status weighted by household size, the HBAI convention) and the income Gini is person-weighted over equivalised HBAI net income. Second, poverty *levels* move with the underlying benefit and uprating parameters across model and data releases; under the pinned releases the budget-neutral changes are −0.23 points BHC and −0.69 points AHC. Third, the wealth-decile results depend on how deciles are constructed in the presence of tied zero-wealth households; we construct equal-weight deciles directly rather than using the model's own decile variable.